Strange Hadron Production in Proton-Proton Collisions: Evaluating PYTHIA 8 Hadronisation Models Against ALICE Measurements

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Abstract

This thesis investigates strange hadron production in proton-proton collisions at the Large Hadron Collider, comparing PYTHIA 8 Monte Carlo tunes to experimental ALICE measurements. The central question examines how effectively junction-enhanced hadronisation (*Junctions*) reproduce observed strange particle production, relative to the baseline model (*Monash*). The analysis examines yields, transverse momentum, baryon-to-meson ratios and angular correlations across different multiplicity percentiles. Junctions demonstrates improvements over Monash, however both models systematically under-predict strange baryon yields relative to the experimental data collected by ALICE, indicating neither model sufficiently explains hadronisation mechanisms at play in proton-proton collisions. This work guides future refinement of Monte Carlo event generators.

Contents

1 Introduction

Particle physics focuses on the study of elementary particles and the forces governing their interactions. These fundamental constituents of matter have existed since the earliest moments of the universe. Modern cosmology points to the universe emerging 13.8 billion years ago, from a hot and dense event known as the Big Bang [1]. In the microseconds following this event, matter did not exist in its current form, but rather as a deconfined plasma containing quarks and gluons [1], known as quark-gluon plasma (QGP) [3, 4]. As the universe expanded and cooled, strong interactions bound quarks together into various colour-neutral combinations. This transition, known as hadronisation, forms matter as we see it today. During this process, protons, neutrons and other hadrons were formed, giving rise to the complex structures populating our universe.

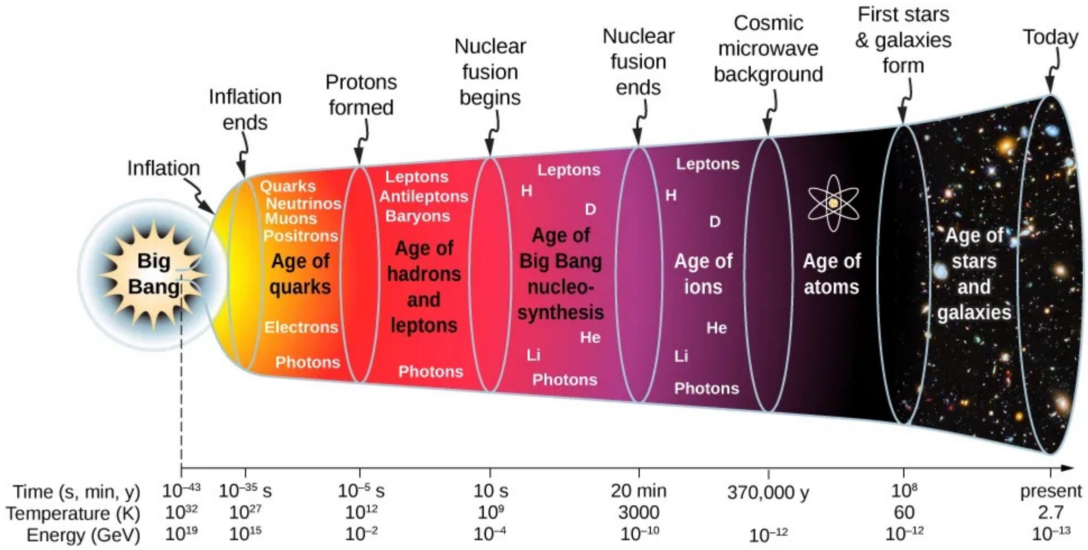


Figure 1: Evolution of the early universe. *Image credit: University Physics [5].*

This sequence of events, from the Big Bang to the formation of stars and galaxies, is summarised in Figure 1. The epoch, referred to as the *Age of hadrons*, represents the first instance of the strong force shaping the structure of matter. Recreating these early-universe conditions is a primary goal of high-energy particle physics. Modern accelerators such as the Large Hadron Collider (LHC) at the European Organization for Nuclear Research (CERN, from the original French *Conseil Européen pour la Recherche Nucléaire*) provide access to collision energies high enough to temporarily deconfine quarks and gluons, allowing us to study these fundamental processes of the early universe at the highest man-made temperatures [6]. Experiments such as A Large Ion Collider Experiment (ALICE) are specifically designed to collide protons and heavy-ions at extreme energies, allowing us to examine how new particles and QGP are formed [7, 8]. Each individual proton-proton collision is referred to as an *event* during this research paper.

Understanding the transition from deconfined matter to hadronic matter is a profound challenge in modern physics. While the general principles are well established, the detailed mechanisms behind hadron formation are still not fully understood. The Standard Model — a set of quantum field theories successfully describing three of the four fundamental forces governing matter — provides the theoretical foundation for our understanding of these interactions [9, 10]. It connects the behaviour of matter under extreme conditions to the subatomic structure of the particles that compose it, offering a bridge between the cosmic and the microscopic.

[1] Quarks and gluons are the building blocks of ordinary matter; they form the protons and neutrons that make up atomic nuclei and thus all visible matter. Quarks and gluons in the proton are collectively referred to as partons [2].

Within the Standard Model, quantum chromodynamics (QCD) is the theory that describes the strong interaction of coloured quarks and gluons, while the electroweak theory describes weak and electromagnetic interactions, leaving only gravity outside its scope [10, 11]. These forces are mediated by carrier particles known as bosons, which transmit interactions between matter particles called fermions. Among these forces, the strong force binds quarks into protons, neutrons, and many types of hadrons [9]. Strange quarks, as members of the quark family, also experience the strong interaction and combine with other quarks to form hadrons. QCD is based on the exchange of mediator particles called gluons, which carry a type of charge known as colour charge. Unlike photons in electromagnetism, which are electrically neutral, gluons themselves carry colour, allowing them to interact not only with quarks but also with one another [9]. The six quark flavours in the Standard Model, along with their charges and masses are summarised in the table below.

Table 1: The six quark flavours in the Standard Model, grouped by generation.

Generation	**Quark**	**Charge**	**Mass (approx.)**
1st	Up (u)	$+\frac{2}{3}e$	2.2 MeV
	Down (d)	$-\frac{1}{3}e$	4.7 MeV
2nd	Charm (c)	$+\frac{2}{3}e$	1.28 GeV
	Strange (s)	$-\frac{1}{3}e$	96 MeV
3rd	Top (t)	$+\frac{2}{3}e$	173 GeV
	Bottom (b)	$-\frac{1}{3}e$	4.18 GeV

As shown in Table 1, the six quark flavours span a wide range of masses. This mass hierarchy is central to understanding how quark flavour affects strong-force interactions at different energies. Asymptotic freedom and confinement are two key features of QCD that describe its behaviour across different energy scales. Asymptotic freedom describes how, at high momentum transfers and short distances, interactions between quarks and gluons weaken, allowing them to behave almost freely. At low momentum transfers and long distances, however, the interactions become increasingly strong, preventing quarks from existing in isolation [4, 12]. The phenomenon of confinement states that quarks and gluons cannot be isolated. Neither quarks nor gluons are observed as free particles, but exist permanently bound in colour-neutral hadrons such as protons and pions [11]. The strength of the strong interaction is characterised by the strong coupling constant, denoted as α_s. Attempting to separate two quarks requires increasing amounts of energy, resulting in the formation of new quark-antiquark pairs [12]. These principles illustrate how QCD behaviour varies with energy scales. At high energies, α_s becomes sufficiently small that perturbative methods (pQCD) can be applied. Perturbation theory expresses calculations as a series expansion in powers of the small coupling constant. However, this approach breaks down at the low-energy regime, where the coupling constant grows large, and the expansion diverges. It is in this non-perturbative regime that phenomena such as confinement and hadronisation occur [4].

Understanding how confinement gives rise to the formation of hadrons is one of the greatest unsolved problems in modern physics. The transition between high- and low-energy regimes presents unique challenges to our understanding of strange particle hadronisation in proton-proton collisions. Investigating strange hadrons in high-energy interactions provides deeper insight into the hadronisation processes and the fundamental mechanisms of QCD.

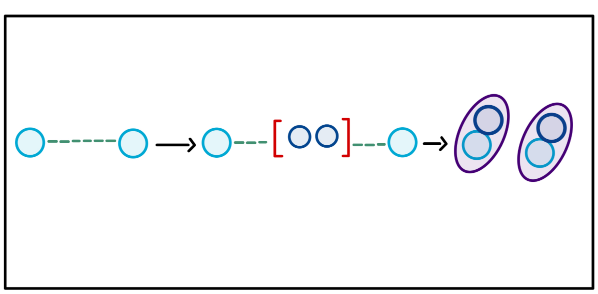
1.1 Hadronisation of strange quarks

Among different quark flavours, strange quarks occupy a unique position within the Standard Model. They are heavier than the up and down quarks, yet significantly lighter than charm and beauty quarks. This intermediate mass gives strange quarks a distinctive role in the study of hadronisation: they are produced less abundantly than the light quarks, but more frequently than the heavy ones. As such, they serve as an ideal probe for studying how quarks form hadrons across different energy regimes. When strange quarks combine with other quarks, they form strange hadrons such as kaons (K), Xi (Ξ) baryons and Omega (Ω) baryons. The intermediate mass of the strange quark also results in the production of strange hadrons being suppressed relative to hadrons containing up and down quarks [13].

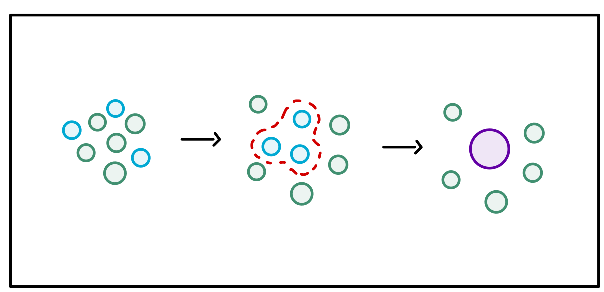
In proton-proton collisions, the system size is very small, requiring local strangeness conservation, which suppresses the relative yield of strange hadrons [4]. In denser systems — such as heavy-ion collisions — this suppression is lifted, leading to strangeness enhancement. This contrast makes strange hadrons useful for testing whether high-multiplicity proton-proton events exhibit behaviour similar to that observed in heavy-ion collisions. Strange baryon-to-meson ratios are particularly informative, as they reflect the underlying hadronisation mechanism. *Fragmentation* processes, where colour strings break to form hadrons tend to favour meson production. *Recombination* or coalescence processes, which become more probable in denser environments, naturally enhance baryon yields [4]. Studying how these baryon-to-meson ratios evolve with event multiplicity allows us to determine whether fragmentation alone can explain strangeness production, or whether recombination mechanisms are operative even in small systems.

Hadronisation is the process by which composite quantum states — hadrons — are formed from their fundamental constituents (quarks and gluons). It marks the point at which the strong interaction becomes too intense for perturbative calculations to remain valid. Consequently, hadronisation is a non-perturbative process that produces a spectrum of mesons and baryons, governed by the principle of *confinement* which prohibits quarks and gluons from existing as free particles [4]. Confinement ensures that the strong force intensifies when quarks are separated, preventing partons from escaping. In high-energy proton-proton collisions, the initial hard scattering process produces high-momentum partons, which subsequently undergo hadronisation. Hadron formation begins in the partonic phase, where only quarks and gluons are present. The transition is characterised by small momentum transfers and strong coupling [4]. The result is the production of mesons and baryons in various colour-neutral combinations of quarks.

Hadronisation is commonly described by two mechanisms: fragmentation and recombination [4]. At high transverse momentum, fragmentation dominates. In this process, a fast moving parton creates a colour string that stretches behind it, then breaks and produces quark-antiquark pairs. These pairs combine with the original parton to form hadrons, leading to jet formation. Naturally, fragmentation tends to produce more mesons than baryons, as it is statistically more likely to form quark-antiquark pairs than three quark combinations. At lower transverse momentum, or denser environments recombination becomes dominant. In this mechanism quarks that are close in phase space can merge directly to form hadrons [4]. For baryons, three nearby quarks recombine, whereas mesons form from quark-antiquark pairs. In events with high particle density, where many partons coexist within a small volume, recombination can lead to an enhancement in baryon production. Proton-proton systems are comparatively small, and as a result, hadronisation is expected to proceed primarily through fragmentation across most of the momentum spectrum. However, in the highest multiplicity proton-proton collisions, where many partons are produced within a confined region, the conditions may resemble those necessary for recombination effects to emerge [14]. Figure 2 illustrates the two main hadronisation mechanisms discussed above. The left panel shows a simplified version of the colour string breaking, the right panel depicts 3 quarks coalescing.



(a) Fragmentation mechanism.



(b) Recombination mechanism.

Figure 2: Different hadronisation mechanisms.

After hadronisation, the newly formed hadrons continue to interact within the hot, dense medium. Inelastic scatterings, altering particle abundance, occur until chemical freeze-out is reached, at which point the particle compositions are fixed. Elastic scatterings, modifying momentum distributions, continue until kinetic freeze out. Subsequently, hadrons stream freely towards the detectors [4].

To describe this complex transformation, physicists use phenomenological models that reproduce features of QCD and aim to depict hadron formation accurately.

1.2 Event generators

Research on hadronisation has advanced through both experimental observations and theoretical modelling. After decades of collider measurements, it has become clear that no perturbative method can adequately describe this complex transition from quarks to hadrons. As a result, phenomenological models have become essential tools for understanding hadronisation.

One of the most successful frameworks for describing hadronisation is the Lund string fragmentation model, which forms the backbone of most modern event generators. In this approach, the colour confinement field is represented as a network of strings that form between partons. As the partons separate, the strings stretch, eventually breaking and producing new quark-antiquark pairs [15]. The probability of such a string breaking is exponentially suppressed with the transverse mass of the produced quarks, as described by the relation:

$$P \sim \exp\left(-\frac{\pi m_T^2}{\kappa}\right) \tag{1}$$

where p_T is the transverse momentum (component of momentum perpendicular to the beam axis - further explored in *Analysis details*), m_T is the transverse mass defined as

$$m_T = \sqrt{p_T^2 + m^2} \tag{2}$$

and κ is the string tension. This exponential behaviour naturally explains why heavier flavours, such as strange quarks, are produced less frequently than the lighter ones [16, 17].

The Lund string model extends this qualitative picture of fragmentation into a mathematical formalism. Its strength lies in its ability to encode confinement and hadron formation within a computational framework suitable for large-scale event simulation. The model has been implemented extensively in Monte Carlo event generators — most notably PYTHIA, one of the most widely used tools in high-energy physics [18]. The string fragmentation picture shown in Figure 3 illustrates the dynamics of hadron formation as described by the Lund string model.

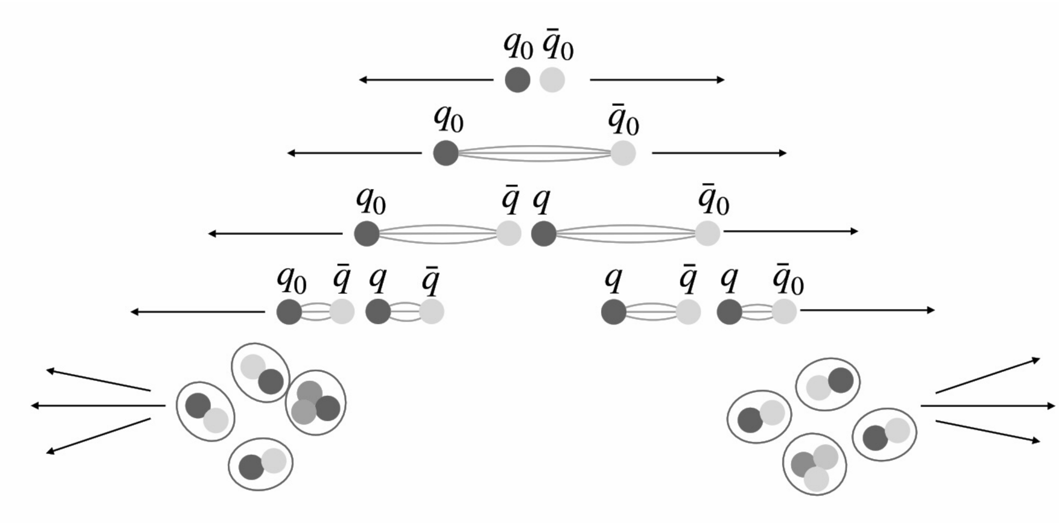


Figure 3: Hadronisation in the Lund string model (i) The initial quarks move apart. (ii) A colour string forms between them as the colour field stretches (iii) The string breaks, creating new quark-antiquark pairs. (iv) Subsequent string breaks continue the process (v) Resulting quarks combine to form colour-neutral hadrons that emerge as jets *Image credit: The European Physical Journal C [19].*

PYTHIA simulates complete particle collisions, such as proton-proton collisions occurring at the LHC. It models every stage of the event, from the initial hard parton scatterings to the final hadronic states, ensuring colour neutrality and confinement are maintained. Over more than three decades of development, PYTHIA has been continuously validated and tuned to achieve agreement with experimental data. The default configuration of PYTHIA, known as Monash 2013, was primarily calibrated using e^+e^- collision data from the Large Electron-Positron Collider (LEP) at CERN and the SLD experiment at SLAC, and later refined using LHC measurements [20]. While the Monash tune successfully reproduces many global observables successfully, there are still shortcomings. PYTHIA tends to significantly underestimate the yield of baryons, an expected consequence of the Lund string model that only allows strings to break and form quark-antiquark pairs, highlighting the need for further theoretical refinement [21].

Recent developments within the PYTHIA framework aim to address these discrepancies through the implementation of Colour Reconnection (CR) models. CR describes the rearrangement of colour strings between partons prior to hadronisation. In multi-parton interactions (MPIs), several independent colour strings are formed within the same event, and CR allows these strings to reconnect in a configuration that minimises the total string length of the system. This rearrangement alters the colour topology of the event and can therefore affect the resulting hadron yields. Physically, CR mimics collective behaviour similar to that observed in larger systems and is particularly relevant in high-multiplicity proton-proton collisions where overlapping colour fields are significant [22].

Earlier versions of PYTHIA implemented CR, but this approach continued to underestimate baryon production. To address this, a newer configuration — CR Mode 2, known as Junctions — was introduced. In this extension, reconnections can form Y-shaped topologies in which three colour lines converge at a single point called a string junction, acting as a natural baryon number carrier [15, 18]. When the three arms fragment, at least one baryon is produced, improving agreement with ALICE measurements in proton-proton collision. This provides a natural explanation for the enhanced baryon yields observed in data, as Junctions effectively increases baryon production without invoking recombination models. An illustration of the main CR topologies is shown below in figure 4.

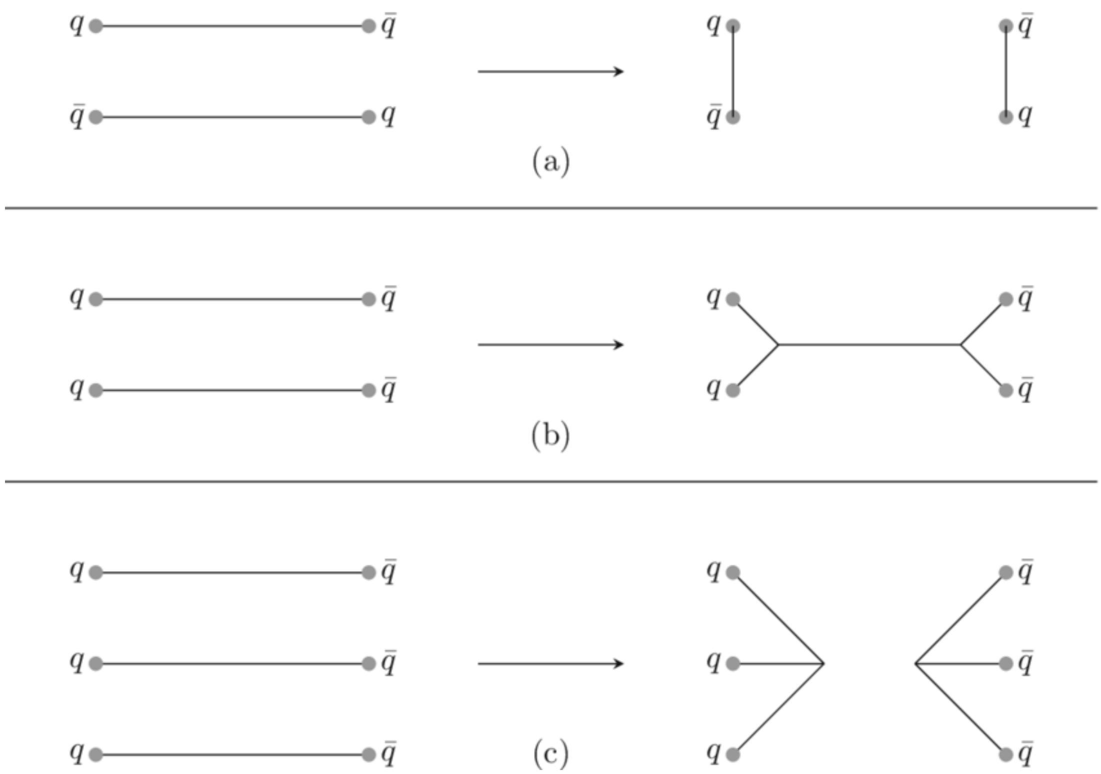


Figure 4: Illustration of CR topologies. For two dipoles, (a) a simple reconnect (b) formation of connected junction-antijunction system. For three dipoles, (c) a disconnected junction-antijunction configuration may form. *Image credit: The European Physical Journal C [23].*

Alternatively, recombination models provide another approach to simulating hadronisation. In this framework quarks proximate in phase space combine directly, rather than through string fragmentation, with quark-antiquark pairs forming mesons and three quarks forming baryons. In dense environments, this becomes the preferred mechanism and naturally enhances baryon production [4]. While not typically expected in proton-proton collisions, it could play a role at very high multiplicities, where parton density becomes significantly large.

Experimentally, hadronisation has been investigated through precise measurements of particle production in proton-proton collisions at the LHC. Experiments such as ALICE have measured the full transverse momentum spectra for mesons and baryons, including strange hadrons, and compared them with PYTHIA predictions [24]. While the generator reproduces the general trends, it often fails to capture the exact magnitude of strange hadron enhancement observed at increasing multiplicities. Such comparisons between theory and data expose several key gaps in our understanding. It remains unclear whether recombination mechanisms operate in high-multiplicity proton-proton collisions, how baryon production can be accurately modelled within existing frameworks, and whether hadronisation is universal across quark flavours or exhibits flavour dependence. Experimental findings by the ALICE collaboration indicate near-universal scaling of strange particle yields with multiplicity of proton-proton collisions, the question remains about whether simulations like PYTHIA, updated with CR Mode 2, can accurately reproduce this behaviour [25].

By addressing these uncertainties the present work contributes to an ongoing effort to test and refine our theoretical understanding of the non-perturbative regime of QCD. Bridging the gap between phenomenological models and experimental observation is essential for developing a consistent and universal description of hadronisation across all systems.

2 Analysis details

This project combines Monte Carlo simulations and comparative data analysis to investigate strange hadron production in proton-proton collisions. The project uses PYTHIA 8, a Monte Carlo event generator that simulates high-energy collisions. Several parameter configurations have been developed, and these optimise agreement between simulated and experimental data, with Monash 2013 being the most widely used [18]. Built over more than three decades of development, the current version of PYTHIA represents the culmination of continuous refinement [26]. The central goal is to understand how strange baryons and mesons are produced during hadronisation. By comparing different hadronisation models, this work aims to determine whether mechanisms beyond string fragmentation and string junctions are necessary to explain the observed baryon enhancement in high-multiplicity proton-proton collisions.

2.1 Observables

To study strange hadron production in proton-proton collisions, several key observables are defined and analysed. These observables provide insight into the mechanisms that govern hadronisation and allow for comparison between simulation and experimental data. Each observable probes a different aspect of the underlying dynamics.

Charged-particle multiplicity (N_{ch}) is one such fundamental observable in high-energy collisions. In this study, it is defined as the number of charged particles produced per event within the pseudorapidity (defined below) range of $|\eta| < 4$ and within transverse momentum of $p_T > 0.15$ GeV [27]. Simulations were generated with $|\eta| < 4$ acceptance to maximise event statistics for rare particles. In the context of PYTHIA, multiplicity is the outcome of MPIs, where several semi-hard scatterings occur in a single collision. These MPIs along with hadronisation contribute to the final number of charged particles. This underscores the importance of understanding N_{ch} for tuning event generators to reproduce complex parton-level interactions and their transition to hadrons. The ALICE Collaboration has shown the dependence of observables on N_{ch} aids in improving the theoretical modelling of the interplay between soft and hard QCD processes governing particle collisions. In this analysis, events are categorised into multiplicity classes expressed as percentiles of the N_{ch} distribution [27].

Another central observable is transverse momentum (p_T), defined as the component of the particles momentum that is perpendicular to the beam axis. It is calculated as:

$$p_T = \sqrt{p_x^2 + p_y^2} \tag{3}$$

The shape of the transverse momentum distribution reflects different processes: low-p_T is associated with soft (non-perturbative) QCD processes where α_s is large, and the opposite for high-p_T. The intermediate p_T range is particularly valuable for understanding strange hadron production, because in this range both fragmentation and recombination can contribute to hadron formation. The combination of two observables, p_T against N_{ch}, provides additional insight. The ALICE collaboration has also shown that as multiplicity increases the mean transverse momentum rises [28].

The analysis also explores baryon-to-meson ratios, such as Λ/K_S^0, calculated within each multiplicity interval. These ratios relate how baryon production scales relative to meson production, and further informs us on the underlying hadronisation mechanism. Meson production typically dominates in most scenarios because statistically forming quark-antiquark pairs is more likely than three quark combinations. Since recombination can also form mesons, a better signature of this mechanism is a relative enhancement in baryon yields relative to mesons. The baryon-to-meson ratio is defined as:

$$R_{B/M} = \frac{N_B}{N_M} \tag{4}$$

where N_B represents the amount of baryons, and N_M the amount of mesons. In addition to baryon-to-meson ratios, studies indicate that the overall yield of strange hadrons increases as a function of multiplicity, a phenomenon referred to as *strangeness enhancement* [29]. Reproducing this remains challenging for current event generators.

Additionally, two particle angular correlations play a key role in this study. For each event the azimuthal angle (ϕ) and pseudorapidity (η) of a particle are recorded, allowing the calculation of their angular separation. Pseudorapidity is defined in terms of the polar angle (θ) — the angle between the particle's momentum and the beam axis — as:

$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right] \tag{5}$$

The azimuthal angle, in turn, describes the particle's position around the beam axis within the transverse plane. The relative azimuthal angle between two particles is given by:

$$\Delta\phi = \phi_1 - \phi_2 \tag{6}$$

The joint two-dimensional distribution of these variables, provides invaluable information on how these particles are spatially correlated and reveals characteristic patterns, further aiding our investigation into hadron production [30].

To ensure meaningful comparison across simulations, the correlation function is normalised per trigger particle, according to the formula:

$$C(\Delta\eta, \Delta\phi) = \frac{1}{N_{\mathrm{trig}}}\frac{d^2N}{d(\Delta\eta)\,d(\Delta\phi)} \tag{7}$$

Where N_{trig} is the total amount of trigger particles in the sample, and N the number of associated particle pairs. This normalisation approach is standard in two-particle correlation analyses. It allows us to directly compare between different PYTHIA tunes or multiplicity classes, despite differences in amounts of events or particle yields [30].

2.2 Simulation and Analysis procedure

All proton-proton collision events were generated using PYTHIA 8.3, a Monte Carlo generator that models the complete evolution of a high-energy collision — from the initial parton scatterings to final hadronic states [18]. PYTHIA implements the Lund string fragmentation model, in which confinement is represented through linear colour strings between quarks and gluons, to simulate the hadronisation process. Two configurations (or *tunes*) were considered in this analysis. The baseline tune of the simulation, Monash 2013, is well-calibrated but it is known to underestimates baryon yields, due to the limitations of standard string fragmentation. Therefore an alternative configuration, Junctions, was also considered. Comparing these two tunes provides insight into whether topological changes to the string field better reproduce strange hadron yields observed at ALICE.

Each simulation was configured to generate proton-proton collisions at a centre-of-mass energy of $\sqrt{s}$ = 7 TeV, to match the ALICE experimental measurements of strange hadron yields. To ensure adequate statistical precision, two hundred million events were simulated and divided into smaller subsamples for analysis. The large-scale computational demands of generating this amount of events, allowed for the use of the NIKHEF computer cluster, where simulations were submitted to the Condor job system. The analysis used ROOT to store and process PYTHIA event data, the standard data analysis framework developed at CERN [31]. Strange hadrons were identified in the simulated output using their Particle Data Group (PDG) codes, as listed in table 2. The analysis focuses on various strange mesons and baryons, including pions ($\pi^{\pm}$) and protons ($p/\bar{p}$) as reference particles (S = 0) for quantifying strangeness enhancement, as detailed in Table 2.

Table 2: List of particles included in the analysis.

Category	**Particle**	**Symbol**	**PDG ID**	**Quark Content**	**Strangeness**
Reference Particles (Non-Strange)					
Mesons	Pion	$\pi^{\pm}$	211 / -211	$u\bar{d}$ / $\bar{u}d$	S = 0
Baryons	Proton	p / $\bar{p}$	2212 / -2212	uud / $\bar{u}\bar{u}\bar{d}$	S = 0
Strange Mesons					
Mesons	Kaon (charged)	$K^{\pm}$	321 / -321	$u\bar{s}$ / $\bar{u}s$	S = ±1
	Kaon (neutral, short)	K^0_S	310	$(d\bar{s} - \bar{d}s)/\sqrt{2}$	S = ±1
Strange Baryons					
Baryons	Lambda	Λ / $\bar{\Lambda}$	3122 / -3122	uds / $\bar{u}\bar{d}\bar{s}$	S = -1
Baryons	Xi (cascade)	Ξ^- / $\bar{\Xi}^+$	3312 / -3312	dss / $\bar{d}\bar{s}\bar{s}$	S = -2
Baryons	Omega	Ω^- / $\bar{\Omega}^+$	3334 / -3334	sss / $\bar{s}\bar{s}\bar{s}$	S = -3

Furthermore, we filtered particles by status code, ensuring only final-state hadrons originating from hadronisation remain. For each event the analysis script loops through all hadrons, assessing their PDG codes and simultaneously recording kinematic properties and identification information such as p_T, η and ϕ. These quantities are then stored in different histogram types.

The multiplicity distribution was then extracted from the stored histograms, establishing the multiplicity percentile classes used in subsequent phases of the analysis. Second, the transverse momentum spectra were produced for each particle within each multiplicity class, allowing for quantitative comparison. Integrated yields were calculated by summing the p_T spectra across all transverse momentum bins for each particle species, and compared to ALICE experimental data. Subsequently, baryon-to-meson ratios were calculated as a function of p_T. Angular correlations between two particles further inform on the dynamics behind hadronisation, and comprises the final section of the analysis. For this analysis, a strange baryon was selected as trigger particle, serving as reference. Relative to the trigger, an associated particle was selected. In this case, both particles were charged Xi (Ξ) particles and were identified by their PDG code (Ξ^-, PDG ID = 3312, and Ξ^+, PDG ID = –3312). To suppress random background contributions, same sign pairs ($\Xi^+\Xi^+$ and $\Xi^-\Xi^-$) are subtracted from opposite sign pairs ($\Xi^+\Xi^-$). This subtraction isolates genuine correlation signals, arising from the same colour string or jet, from accidental coincidences between unrelated particles.

A 3D ROOT histogram tracks transverse momentum of the trigger and associate particle, along with their pseudorapidity difference. This plot maps the phase-space coverage of the dataset. To visualize how Ξ baryons are correlated in angular space, two-dimensional histograms are produced showing $\Delta\phi$ versus $\Delta\eta$. After integrating the two-dimensional distribution of $\Delta\eta$, a one-dimensional $\Delta\phi$ function is graphed, to extract key insights from the distribution. Observing a near-side peak at $\Delta\phi = 0$ reflects hadrons formed from the same jet, and an away-side peak at $\Delta\phi = \pi$ indicates back-to-back production. This peak will be identified in our correlation distribution and observed with regards to changing multiplicity [30]. ROOT is used for all processing and visualisation. Custom C++ macros create histograms and plots. The analysis is performed at small scale first to verify the workflow, and hereafter is submitted to the NIKHEF computing cluster to be performed on millions of events.

Due to the large amount of simulated collisions, the subsampling technique is used to estimate statistical uncertainties. The full dataset is divided into ten independent subsets, containing equal amounts of events. Within each subset, observables are calculated independently. The mean value across the subset gives the final observable, hence the standard deviation provides the statistical uncertainty. This approach allows for robust uncertainty calculation, even in large datasets. It is especially suited for Monte Carlo studies where fluctuations arise at random.

2.3 Expected outcomes and ethics

This analysis is expected to demonstrate that junction-enhanced configurations of PYTHIA reproduce strangeness enhancement trends better, compared to the default Monash tune. The findings will contribute to the scientific community by broadening our understanding of hadronisation in proton-proton collisions, with particular focus on strange particle production. Hadronisation as a part of the non-perturbative regime and a key component of QCD, remains a challenge to our understanding. By analysing strange hadron production, this research will provide insights into how models like PYTHIA currently succeed or require improvements.

The results will have direct practical implications for the fine-tuning of Monte Carlo event generators, which are indispensable tools in particle physics. Tools such as PYTHIA revolutionize modern particle physics, allowing us to theoretically evaluate experimental results. Improved modelling of strange hadron formation increases the accuracy of simulations used by collaborations worldwide. Ultimately, this thesis bridges the gap between the experimental study and theoretical description of hadronisation. By determining where we currently succeed and fall short in our understanding of hadronisation, the project guides future improvements of Monte Carlo event generators, and our overall understanding of the building blocks of the world around us.

This project uses simulated data, produced through the Monte Carlo simulator PYTHIA, as well as experimental results available through the ALICE Collaboration. This data is freely available so there are no confidentiality concerns. The results of this research will similarly, also be made available, to ensure transparency in the scientific community and allowing others to build upon this work. The computational resources used for data analysis in this project, provided by NIKHEF, will be used efficiently and responsibly ensuring enough resources are available to the wider research community and a small environmental footprint is left.

3 Results

3.1 Characterising the events

The charged-particle multiplicity distribution provides the fundamental observable of this research. All other observables are presented as a function of charged-particle multiplicity, making it indispensable in this analysis. Figure 5 shows the multiplicity distribution N_{ch} for all events generated with the Monash tune in PYTHIA. The distribution is presumed to be identical to the Junctions distribution, as both use the same event generation, but differences arise in the hadronisation process.

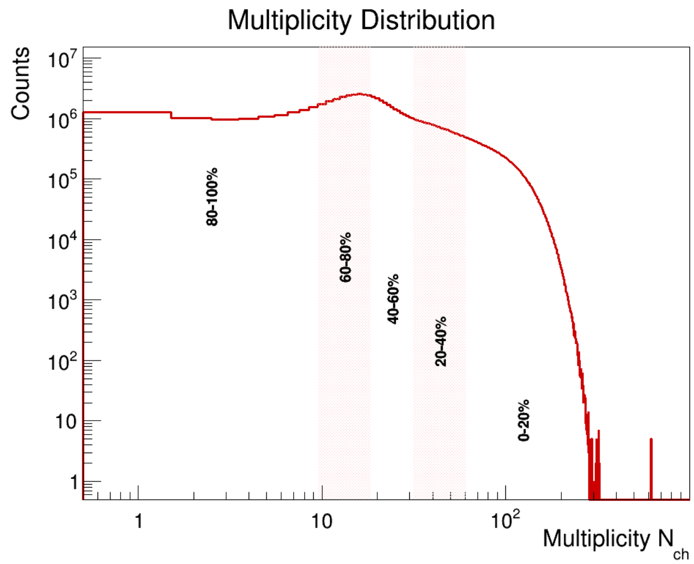
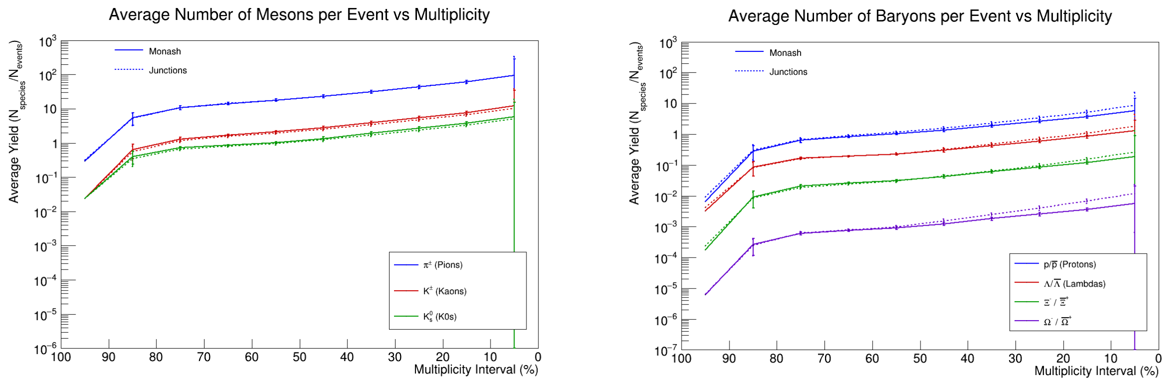


Figure 5: Charged-particle multiplicity distribution. The distribution spans six orders of magnitude, with the multiplicity percentile bins overlaid. The multiplicity is defined by $p_T > 0.15$ GeV/c and $|\eta| < 4.0$. This binning is used for all subsequent yield analyses.

The multiplicity distribution is constructed by counting all charged particles in each event with $p_T > 0.15$ GeV/c within the pseudorapidity window $|\eta| < 4.0$. Events are sorted into multiplicity percentile classes defined as cumulative integrals of this distribution. The binning strategy ensures fair comparison, with approximately equal event statistics across bins while enabling the study of strange hadron production across the full collision energy range. The spectrum falls steeply over six orders of magnitude, with most events populating the low to intermediate N_{ch} region and only a small fraction reaching the high-multiplicity regime. Importantly, ALICE experimental measurements span ten multiplicity classes covering the full 0-100% cross-section, providing sufficient overlap for model-to-data comparison across the entire PYTHIA percentile distribution.

The average number of particles produced per event within each multiplicity provides insight into how particle production scales with event activity. Figure 6 shows the per-event yields for all identified particle species (π, K, p, Λ, Ξ, Ω) as a function of multiplicity percentile for both Monash and Junctions tunes.



(a) Average meson yields: π , K, K^0_S.

(b) Average baryon yields: p, Λ, Ξ, Ω.

Figure 6: Average yields vs. multiplicity for mesons (left) and baryons (right).

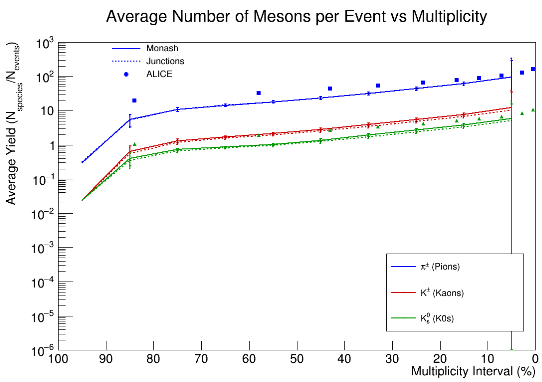
Averaging the per-event yields within each multiplicity, a clear rise is observed in each hadron species curve. Pions (π) are the most abundant, followed by Kaons (K), Protons (p), Lambdas (Λ), Xis (Ξ), and Omegas (Ω), consistent with increasing strangeness content. The enhancement factor is defined as the ratio of the yield slope for Junctions relative to Monash, quantifying how much more strongly a particle scales with event multiplicity under the Junctions tune, and is explored in Table 3 below.

Table 3: Yield slopes as a function of multiplicity for Monash and Junctions tunes.

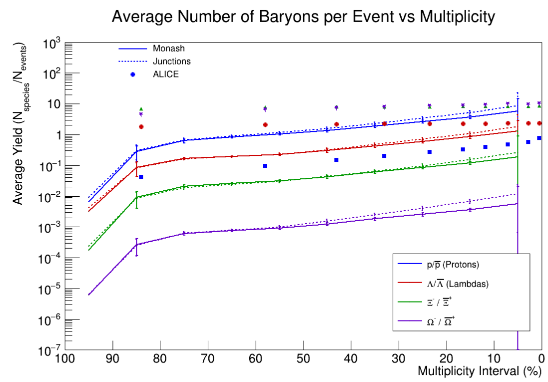
Particle	**Monash**		**Junctions**		**Enhancement**
	Slope	**Error**	**Slope**	**Error**	**Factor**
Mesons					
$\pi^{\pm}$ (Pions)	0.891	0.136	0.873	0.130	0.98
$K^{\pm}$ (Kaons)	0.113	0.018	0.098	0.015	0.86
K^0_s (K^0_s)	0.054	0.009	0.047	0.007	0.87
Baryons					
$p/\bar{p}$ (Protons)	0.054	0.008	0.080	0.015	1.47
$\Lambda/\bar{\Lambda}$ (Lambdas)	0.012	0.002	0.016	0.003	1.34
Ξ^- / $\bar{\Xi}^+$ (Xis)	1.8×10^{-3}	2.8×10^{-4}	2.3×10^{-3}	4.6×10^{-4}	1.33
Ω^- / $\bar{\Omega}^+$ (Omegas)	5.3×10^{-5}	0.9×10^{-5}	1.0×10^{-4}	0.2×10^{-4}	2.00

Comparing tunes at fixed percentiles, Junctions tends to sit slightly below Monash for meson yields and slightly above for baryon yields. The slightly suppressed meson yields are expected: Junctions' CR mechanisms preferentially create baryons due to the junctions topology. The enhancement factor of mesons (0.86-0.98) demonstrates that Junctions produces 2-14% fewer mesons at fixed multiplicity percentiles. Differences intensify at higher multiplicities where MPIs become abundant enough for junction topology formation to compete effectively with string fragmentation. Table 3 reveals strong strangeness dependence: π exhibit the steepest multiplicity slope (0.891), indicating that abundant light particles scale strongly with event activity, while heavier and more exotic strange baryons show progressively weaker slopes ($p = 0.05$, $\Lambda = 0.01$, $\Xi = 0.002$, and $\Omega = 0.00005$). The Ω production demonstrates the same linear multiplicity scaling as lighter baryons, yet begins from such a suppressed baseline (due to scarcity) that absolute yield scaling remains nearly imperceptible. Baryon enhancement factors consistently exceed unity, corroborating that junction topologies preferentially enhance baryon production. This extends to ultra-rare strange baryons, like Ω that exhibits an enhancement factor of 2, indicating that junction mechanisms are twice as efficient at producing strange baryons. This suggests that in the Junctions tune the strangeness limitation becomes relatively less severe for ultra-rare species. However, the small absolute yield of Ω means this twofold enhancement manifests only as a tiny slope change, indicating models cannot overcome the fundamental scarcity of strangeness in proton-proton collisions.

To validate the PYTHIA simulations against real experimental measurements, the predictions from both Monash and Junctions are directly compared with the ALICE collaboration data from proton-proton collisions at 7 TeV taken at the LHC. A critical consideration is the different η acceptance windows: PYTHIA uses $|\eta| < 4$ while ALICE uses $|\eta| < 0.5$, resulting in an acceptance factor (representing the ratio of η ranges covered) of 8. Despite the η differences, yields can be compared meaningfully through analysis macros accounting for acceptance-corrected yields. The underlying assumption is that particle production is approximately flat in η across the mid-η region, allowing the ALICE measurements to represent an eighth of the PYTHIA multiplicity spectrum. Deviations from perfect flatness introduce systematic uncertainties in this model-to-data comparison.



(a) Average meson yields over pions compared to ALICE.



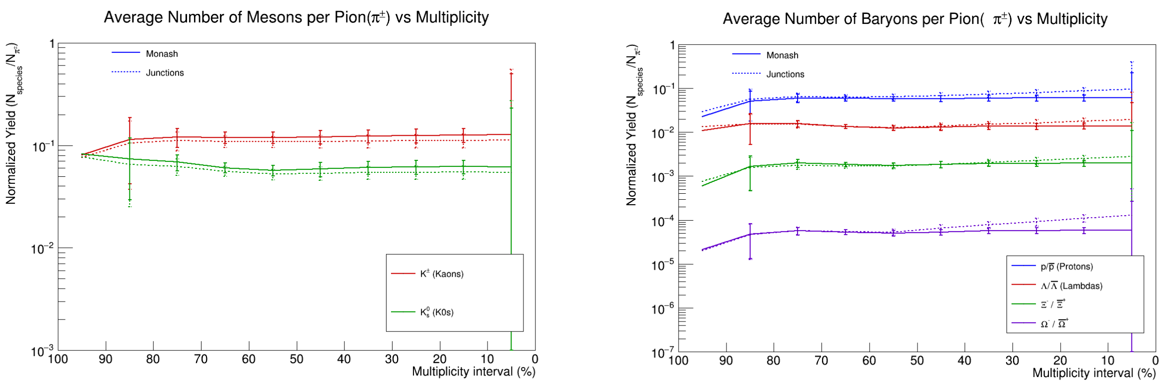
(b) Average baryon yields compared to ALICE.

Figure 7: Average yields vs. multiplicity for mesons (left) and baryons (right) compared to ALICE.

Both Monash and Junctions under-predict meson yields relative to ALICE experimental measurements. ALICE data lies systematically above model predictions across all measured multiplicities, though the discrepancy slightly decreases at higher multiplicities. Monash and Junctions show near identical meson prediction, especially for π, pointing towards tune-independence. This indicates meson under-prediction does not stem from hadronisation differences between tunes, but rather shared issues across both models. The discrepancies reflect inherent limitations of the Lund string fragmentation framework, with neither Monash nor Junctions capturing the meson yield perfectly. All strange baryon species are under-predicted relative to ALICE by both models, with discrepancies increasing with strangeness content. In contrast, p yields are over-predicted by the models, likely because protons are produced abundantly through string fragmentation from the abundant up and down quarks, and are not constrained by strangeness activity. The 0-20% bin exhibits anomalously large error bars possibly reflecting statistical limitations in this extreme multiplicity region where event counts decrease sharply. Both models show similar predictions at low multiplicity but diverge in the high-multiplicity region where Junctions shows progressive enhancement relative to Monash, though both remain substantially below ALICE measurements for strange baryons.

The Model/ALICE ratio plots quantify agreement across the measured multiplicity range and can be found in Appendix A. Ratios at 1 indicate perfect agreement, values below 1 indicate model under-prediction. The K^0_s ratio plot, Figure 26, shows both Monash and Junctions clustering around 0.50-0.80, reinstating that both models under-predict strange K yields. Across the entire multiplicity spectrum Monash achieves better agreement with ALICE than Junctions, for K^0_s. For Λ, Ξ and Ω, this ratio drastically drops to 0.0-0.5 indicating dramatically worse agreement for both models, and indicating fundamental deficiencies in both models' treatment of strange baryon production. This suggests that neither string fragmentation nor the junctions-enhanced approach fully captures the mechanisms generating strange baryons in high-energy proton-proton collisions. Nevertheless, all strange baryon Model/ALICE plots show Junctions sitting above Monash, confirming that the junction-enhanced approach produces higher yields closer to experimental values than standard string fragmentation.

Different collisions produce different numbers of total particles. By expressing yields as ratios to π, one can see which particles are relatively enhanced or suppressed, independent of overall collision multiplicity.



(a) Average meson yields over pions: K, K^0_S.

(b) Average baryon yields over pions:p, Λ, Ξ, Ω.

Figure 8: Average yields-to-π vs. multiplicity for mesons (left) and baryons (right).

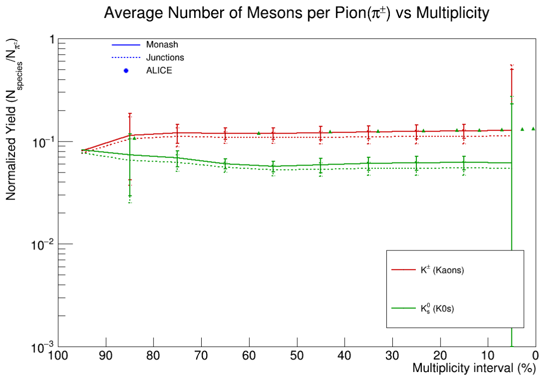
Both the K/π and K^0_S/π ratios remain approximately flat across multiplicity, indicating that K production scales proportionally with π production across all collision sizes in proton-proton collisions. Minor differences between models (slightly steeper K slope in Monash, while K^0_s shows stronger suppression in Junctions) are negligible compared to the pronounced baryon enhancements. Most baryons show upward trends with multiplicity, with the exception of Λ in Monash, which exhibits suppression. All baryon enhancements intensify in the 0-20% bin. The Junctions tune consistently produces higher baryon yields, but the effect strengthens at high multiplicities where junction topologies become most competitive with fragmentation.

Table 4: Particle-to-pion yield slopes as a function of multiplicity.

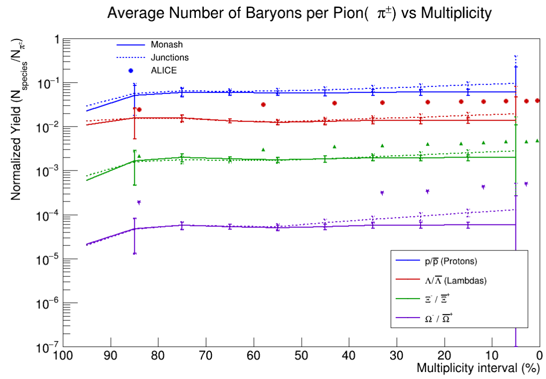
Particle / Pion	Monash		Junctions		Enhancement Factor
	Slope	Error	Slope	Error	
Mesons					
$K^{\pm}$ (Kaons)	1.3×10^{-4}	3.7×10^{-4}	4.0×10^{-5}	3.2×10^{-4}	0.31
K^0_s (K^0_s)	-1.3×10^{-5}	1.7×10^{-4}	-4.9×10^{-5}	1.5×10^{-4}	3.71
Baryons					
$p/\bar{p}$ (Protons)	5.1×10^{-5}	1.8×10^{-4}	4.0×10^{-4}	2.3×10^{-4}	7.87
$\Lambda/\bar{\Lambda}$ (Lambdas)	-3.7×10^{-6}	3.8×10^{-5}	5.8×10^{-5}	4.6×10^{-5}	-15.5
Ξ^{-} / $\bar{\Xi}^{+}$ (Xis)	2.5×10^{-6}	5.5×10^{-6}	1.3×10^{-5}	6.5×10^{-6}	5.06
Ω^{-} / $\bar{\Omega}^{+}$ (Omegas)	9.9×10^{-8}	1.7×10^{-7}	7.8×10^{-7}	2.6×10^{-7}	7.85

Table 4 confirms near zero slopes for K, establishing that K scales proportionally with π. Mesons show no significant multiplicity dependence.
The p/π and Ω/π ratios show the strongest enhancements, while the Xi/π ratio shows moderate enhancement. For p, Junctions shows 8 times stronger baryon enhancement than Monash. The Λ/π slope notably shows a sign reversal: Monash predicts suppression while Junctions shows enhancement. The sign flip suggests that a junction-enhanced approach substantially alters production patterns of the single-strange baryon. The Ξ/π and Ω/π ratios show an enhancement of 5.06 and 7.85 respectively, demonstrating that Junctions enhances even the rarest strange baryons.

These PYTHIA predictions are also compared with measurements from the ALICE collaboration at the LHC.



(a) Average meson yields over pions compared to ALICE.



(b) Average baryon yields over Pions compared to ALICE.

Figure 9: Average yields-to-π vs. multiplicity for mesons (left) and baryons (right) compared to ALICE.

Both Monash and Junctions under-predict particle-to-π ratios relative to ALICE measurements. For mesons, particularly K^0_s, both models under-predict by a relatively uniform factor, with Monash still achieving slightly better agreement. The baryon-to-π ratios present a clear diagnostic: ALICE measurements exceed model predictions, but Junctions consistently produces higher ratios. Neither Monash nor Junctions yet adequately explain observed abundances relative to π. Junctions — designed to enhance baryon production through CR — still under-predicts ALICE strange baryons, suggesting that topology improvements alone are insufficient.

The Model/ALICE ratio plots of the yield-to-π ratios can be found in Appendix A. This confirms that both models severely under-predict particle-to-π ratios, yet Monash achieves better agreement for mesons and Junctions achieves better agreement for baryons at high-multiplicity. Critically, examining particle-to-π yields independent of overall yield reveal that the conclusions established earlier in the analysis persist: Monash achieves better agreement with ALICE for meson production, while Junctions demonstrates a clear improvement for baryon yields, particularly at high multiplicities where junction topologies become dominant. While models are moving in the right direction, they remain insufficient to explain the full strangeness enhancement pattern observed experimentally.

3.2 Transverse momentum spectrum

Figure 10 presents a two-dimensional density map illustrating the relationship between transverse momentum and event multiplicity. The colour scale, displayed on a logarithmic scale ranging from 1 to 10^7, represents the number of charged particles in each bin, with red indicating the highest density and blue the lowest. The distribution exhibits a characteristic wedge-like shape, with the highest density concentrated at low p_T (0-5 GeV/c) and low multiplicity (1-10). As multiplicity increases, the distribution broadens significantly, extending to progressively higher p_T values. Events with multiplicities above 100 produce particles across a much wider p_T range, reaching up to 50 GeV/c and beyond. This multiplicity-dependent broadening reflects that more complex events undergoing more hard scattering activity, produce higher momentum particles, whereas events at low multiplicity naturally produce soft particles.

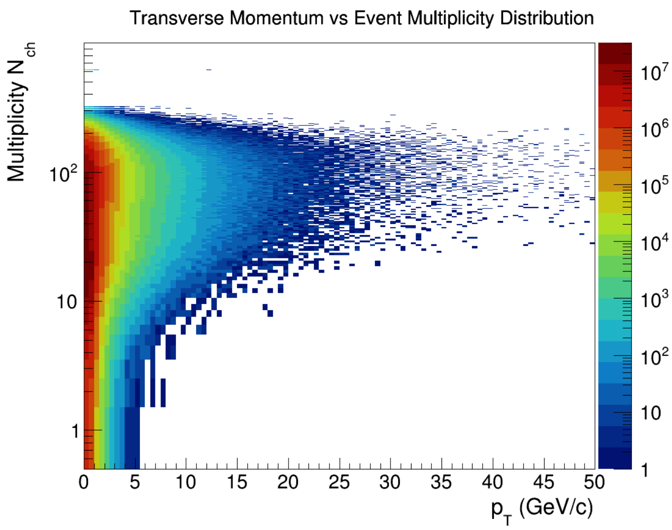


Figure 10: 2D histogram of transverse momentum versus multiplicity.

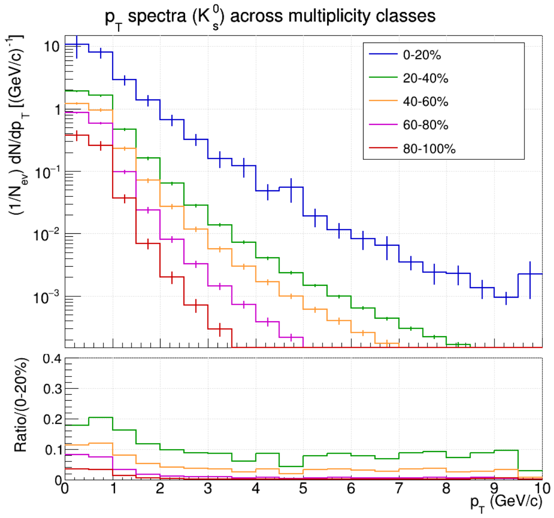
The p_T analysis involves slicing the two-dimensional distribution of p_T versus event multiplicity shown in Figure 10, into one-dimensional histograms for each multiplicity class. This is accomplished by projecting the two-dimensional histograms onto the p_T axis within the multiplicity bin boundaries. For each particle species and hadronisation model, five p_T spectra are extracted corresponding to the five multiplicity classes: 0-20%, 20-40%, 40-60%, 60-80%, and 80-100%. The spectra are normalised to show the differential particle yield as a function of transverse momentum, enabling direct comparison of spectral shapes across multiplicity classes and hadronisation models.

The purpose of examining p_T spectra is twofold:

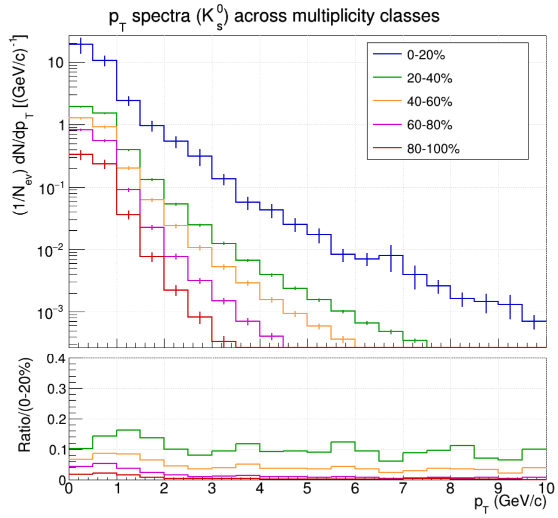
1. Does strangeness content determine whether Junctions shows effects on the p_T spectra?
2. Do Junctions simulated particles show harder p_T spectra or extended tails compared to Monash?

Comprehensive transverse momentum spectra for all measured particle species are presented in Appendix A. The analysis focuses on strange hadrons (K^0_s, Λ, Ξ, Ω) as the primary particles for investigating hadronisation mechanisms, as they are sensitive to strangeness-dependent effects of the PYTHIA models. Light hadrons (p, π, K) and non-strange particles, serve as reference for comparison, demonstrating baseline patterns. Analysis of the reference particle spectra reveals that π and p spectra are nearly identical between models across all multiplicity classes, showing the characteristic exponential-like decay with peaks at approximately 0.0-0.5 GeV/c. Both models show the same hardening trend with increasing multiplicity: higher-multiplicity events produce particles extending to higher p_T values. The spectral shapes remain consistent across multiplicity classes in both models, suggesting that light particle production follows a universal mechanism relatively independent of choice of tune. K spectra show peaks similar to or slightly softer than π, indicating K production follows a pattern consistent with light hadrons.

Consequently, only strange hadrons will be explored in more depth in the following section.



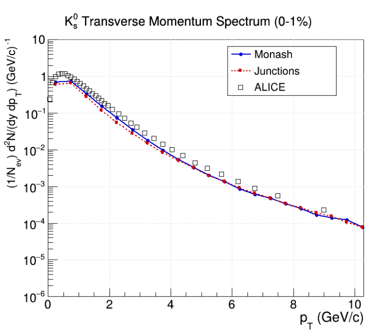
(a) p_T spectra of K_s^0 simulated with Monash.



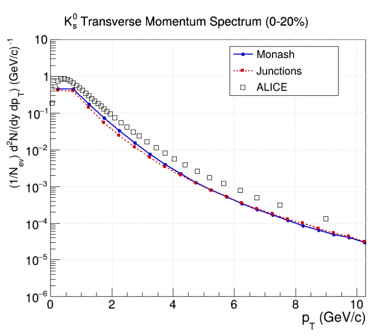
(b) p_T spectra of K_s^0 simulated with Junctions.

Figure 11: p_T spectra of K_s^0 simulated with Monash (left) and Junctions (right).

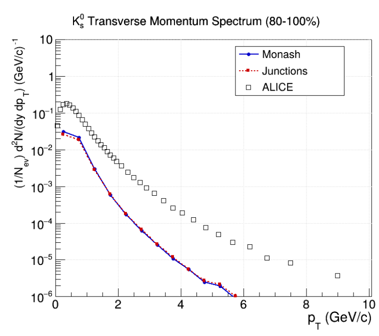
The K_s^0 p_T spectra continues to show clear multiplicity-dependent evolution. Figure 11 shows that K_s^0 spectra in both models display exponential-like decay with a peak occurring in the first bin between 0.0-1.0 GeV/c. A pronounced multiplicity-dependent hardening is evident across all percentiles: low multiplicity (80-100%) events produce soft spectra that drop steeply, while high-multiplicity (0-20%) events produce a broad distribution extending up to 15 GeV/c. Monash and Junctions produce nearly indistinguishable K_s^0 spectra, however, examining the ratio of the multiplicity spectra to the 0-20% spectra, the intermediate multiplicity classes peak marginally higher between 0-2 GeV/c for Monash. However, Junctions shows slightly elevated yields in the very low-p_T region (0-0.5 GeV/c) for the highest multiplicity events (0-20%), suggesting junction topologies enhance soft particle production in dense collision systems.



(a) 0-1%



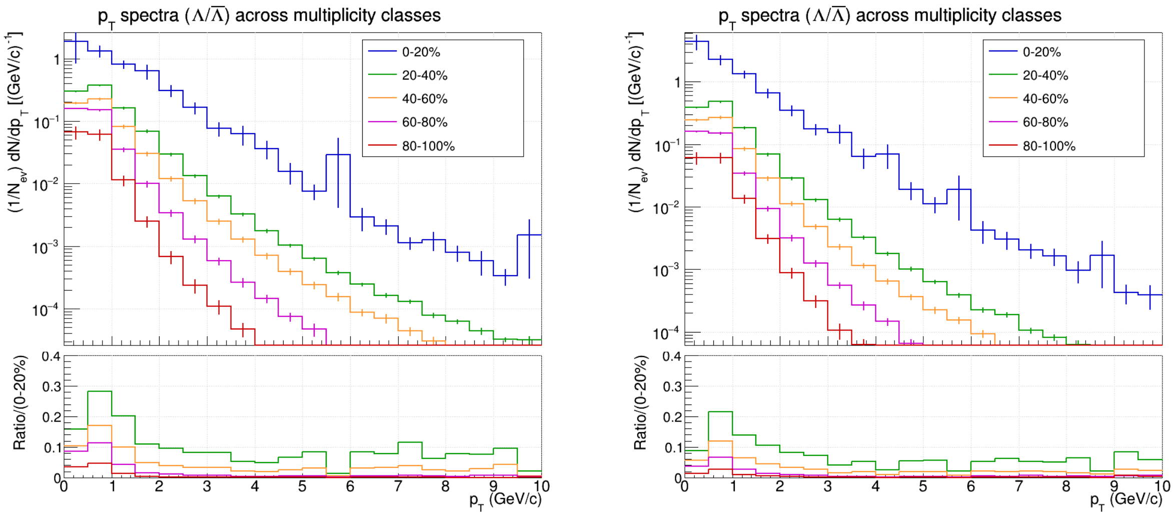
(b) 0-20%



(c) 80-100%

Figure 12: Simulated p_T spectrum of K_s^0 compared to ALICE experimental data.

Comparing to ALICE experimental data, both Monash and Junctions perform well, especially at the highest multiplicities (0-1%). Spectral shapes match ALICE well, both illustrating exponential decay. At the lowest multiplicities both models begin to under-predict yields at high p_T. Monash and Junctions are nearly indistinguishable at low and high p_T, but slightly separate at 1 GeV/$c < p_T <$ 3 GeV/c. This separation in the intermediate p_T region mirrors the marginal differences observed in Figure 11.

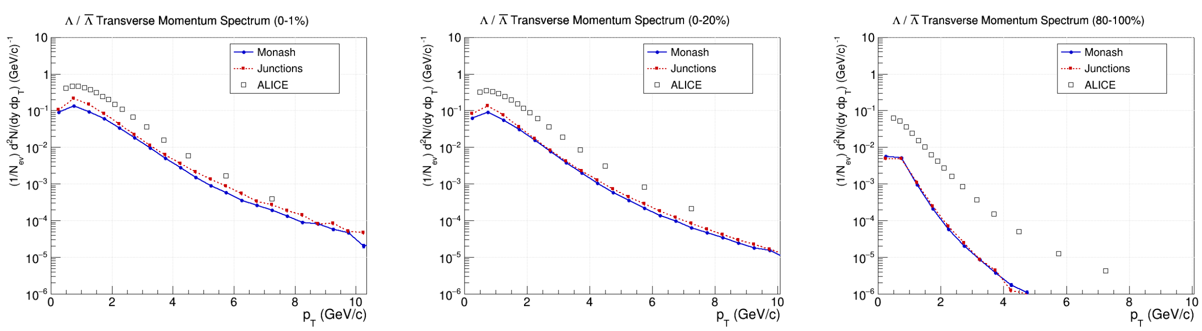


(a) p_T spectra of Λ simulated with Monash.

(b) p_T spectra of Λ simulated with Junctions.

Figure 13: p_T spectra of Λ simulated with Monash (left) and Junctions (right).

The Λ p_T spectra begin to show more substantial variation between the two models. Figure 13 shows exponential-like decay, continuing the trend of peaking in the first p_T bin. While the overall shapes remain similar between Junctions and Monash, the Monash model shows a slightly harder spectrum, particularly evident in the intermediate classes (40-60%). Junctions model shows a softer p_T spectrum, but similar to K_s^0, displays enhanced production in the very low-p_T region (0-0.5 GeV/c) compared to Monash. This low-p_T enhancement combined with the high-p_T suppression indicates a systematic softening of the Λ spectrum under junction-enhanced topologies. This suggests that junction topologies redistribute Λ baryon production toward lower p_T rather than uniformly enhancing production across all kinematic ranges.



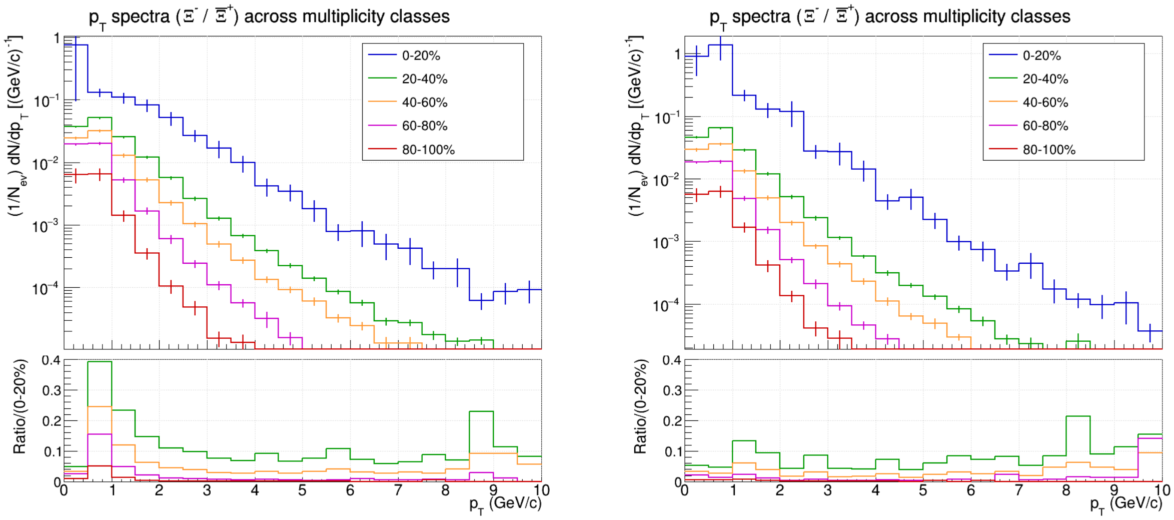
(a) 0-1%

(b) 0-20%

(c) 80-100%

Figure 14: Simulated p_T spectrum of Λ compared to ALICE experimental data.

Comparing to ALICE experimental data, both models show best agreement at the highest multiplicities. At lower multiplicities, model predictions continue to diverge more significantly from ALICE values. The wider dispersion compared to the K_s^0 spectrum suggests that Λ production is more sensitive to hadronisation mechanisms and strangeness effects. This could indicate that strangeness content does determine junction topology effects.

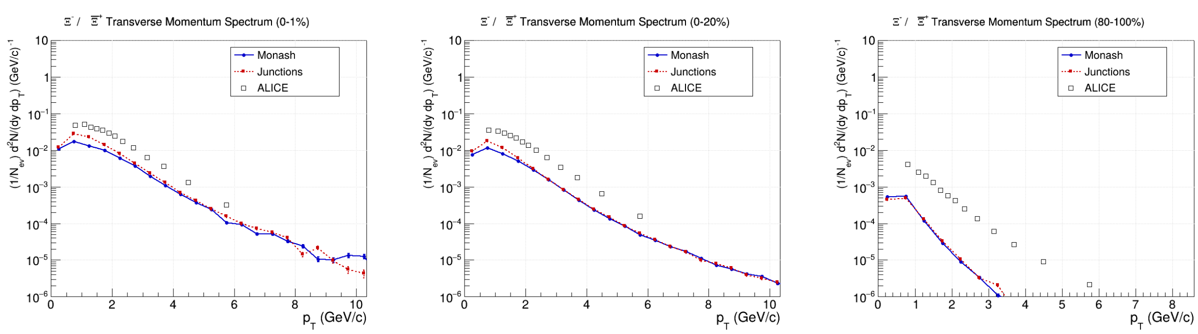


(a) p_T spectra of Ξ simulated with Monash.

(b) p_T spectra of Ξ simulated with Junctions.

Figure 15: p_T spectra of Ξ simulated with Monash (left) and Junctions (right).

The Ξ p_T spectra continue to reveal strangeness-dependent hadronisation differences. Figure 15 shows both Monash and Junctions display exponential-like decay, but with the first noticeable difference in peak positions. The Monash model peaks in the first bin at 0-0.5 GeV/c with a yield below 1, and the Junctions model in the second at 0.5-1 GeV/c with a yield above 1. This peak shift indicates Junctions produces doubly-strange baryons at higher transverse momenta compared to the baseline fragmentation model. Intermediate classes show harder spectra and extended p_T tails in Monash. Multiplicity-dependent hardening continues to be a significant feature of the p_T spectra graphs analysed.



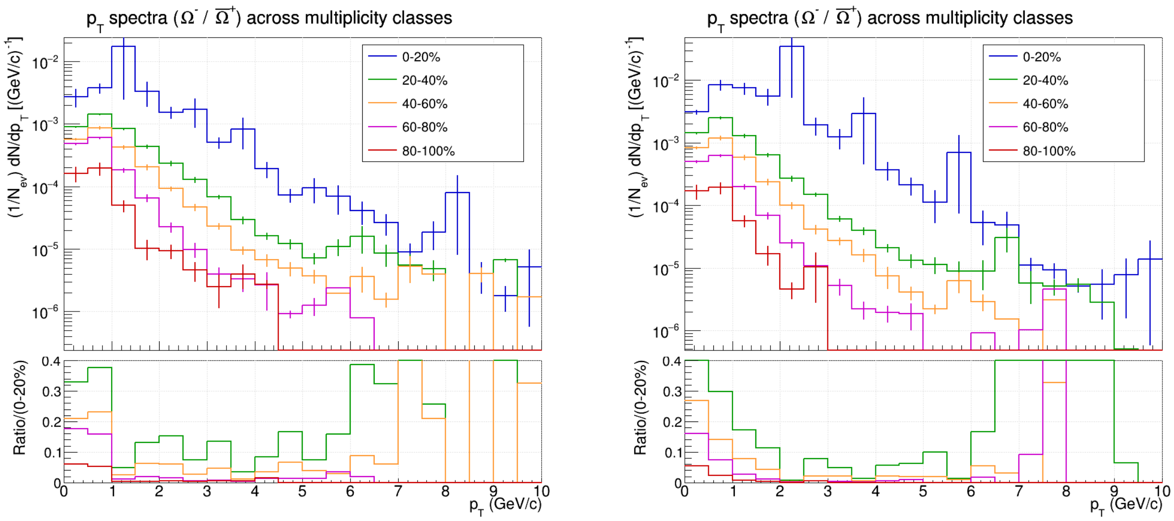
(a) 0-1%

(b) 0-20%

(c) 80-100%

Figure 16: Simulated p_T spectrum of Ξ compared to ALICE experimental data.

Comparing to ALICE experimental data, both models show close alignment to ALICE at high multiplicities (0-20%). At low multiplicities (80-100%) both models diverge significantly from experimental data, but converge closely with each other. This indicates systematic deficiencies in reproducing Ξ particles in quieter production environments. At high multiplicities and low-p_T (<2 GeV/c), the Junctions model sits slightly above Monash, suggesting junction topologies better reproduce Ξ particles in extreme collision conditions. As noted, models converge again at higher p_T, indicating junction effects on double-strange baryons are kinematically sensitive, with topology enhancement most pronounced in soft-p_T regions of high-multiplicity collisions.



(a) p_T spectra of Ω simulated with Monash.

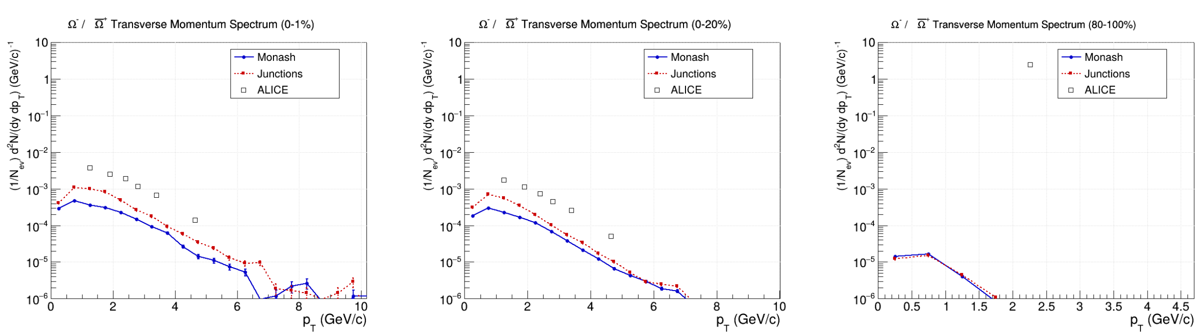
(b) p_T spectra of Ω simulated with Junctions.

Figure 17: p_T spectra of Ω simulated with Monash (left) and Junctions (right).

The Monash and Junctions Ω p_T spectra exhibit fine structure—multiple peaks and fluctuations in the distribution—particularly visible in the high-multiplicity classes. The Ω spectra reflect the extreme rarity of triple-strange baryons in both simulations, resulting in limited statistics and substantial fluctuations. Despite this, Junctions produces higher yields across the measured p_T range, consistent with earlier observed patterns.

Notably, similar to the Ξ analysis, the Ω spectra shows a peak shift between models. The peak in Monash occurs at 1-1.5 GeV/c, while the Junctions model peaks at 2-2.5 GeV/c. This peak shift is even more prominent for Ω than Ξ, suggesting junction-enhanced mechanisms increasingly redistribute rarer strange baryons towards higher-p_T. This is strangeness-dependent kinematic redistribution, indicating Junctions do not simply boost baryon production but fundamentally alter the momentum distribution in ways sensitive to quark flavour. These findings compliments the earlier determined strangeness dependent yield enhancement.

Sparse statistics limit detailed spectral interpretation, but the consistent pattern of kinetic redistribution is noteworthy.



(a) 0-1%

(b) 0-20%

(c) 80-100%

Figure 18: Simulated p_T spectrum of Ω compared to ALICE experimental data.

Comparing to ALICE experimental data, Junctions sits above Monash across the entire multiplicity range, particularly evident at high-multiplicity and soft-p_T. At low-multiplicity both models show extremely limited statistics, making detailed comparison difficult. Junctions consistently produces higher Ω yields than Monash across all multiplicities, indicating that junction topologies enhance rare particle yields.

3.3 Baryon-to-Meson ratios

Baryon-to-meson ratios are a powerful tool to analyse hadronisation mechanisms by isolating the relative production of three-quark systems against two-quark systems, independent of overall particle yield. The baryon-to-meson ratio eliminates common systematic uncertainties, allowing for direct comparison of how different models preferentially produce one particle type over another. In the Lund string fragmentation model, represented by Monash, quark-antiquark pairs are created preferentially. Junction-enhanced systems, represented by the Junctions tune, introduce CR mechanisms that form three-quark topologies, creating additional pathways for baryon production. If junctions mechanisms are effective, they should manifest as enhanced baryon-to-meson ratios, particularly at high-multiplicity, with more MPIs and possibilities of recombination. Both Monash and Junctions will produce ratios substantially less than unity across the measured p_T range. This sub-unity ratio reflects the fact that mesons are universally favoured over baryons because they are lighter and require fewer constituents. Meson production can occur through both fragmentation and recombination, and similarly baryon production can also proceed through both pathways. However, the underlying preference for meson production always dominates. The most telling factor is the relative enhancement: if Junctions is truly enhancing baryon production, it should produce higher ratios than Monash. Moreover, the shape of the baryon-to-meson curve informs of whether the baryon-enhancement scales with event complexity. By examining how the ratio evolves over the multiplicity percentiles, one can determine whether junction mechanisms operate uniformly across all multiplicity classes.

To effectively demonstrate the multiplicity-dependent behaviour of baryon-to-meson ratios while maintaining clarity, the analysis will focus on the two most extreme multiplicity percentiles: the highest-multiplicity bin (0-20%) and the lowest-multiplicity bin (80-100%). The 0-20% bin represents dense, high-activity events, while the 80-100% bin represents more isolated systems.

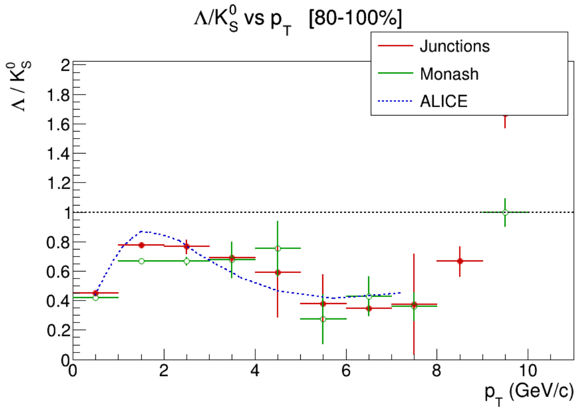
Specifically, we calculate the baryon-to-meson ratio as:

$$R_{\text{baryon/meson}} = \frac{N_{\text{baryon}}(p_T)}{N_{\text{meson}}(p_T)} \tag{8}$$

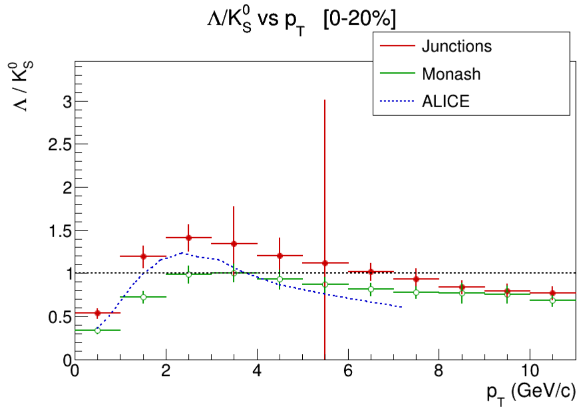
We focus on three representative ratios:

- Λ/K^0_S (strange baryon to strange meson): Tests strange quark hadronisation
- Ξ/K^0_S (double-strange baryon to strange meson): Tests two-strange-quark production
- Ω/K^0_S (triple-strange baryon to strange meson): Tests three-strange-quark production

This strangeness progression allows us to see whether Junctions effectiveness depends on quark flavour. These ratios are computed in the p_T range 1-11 GeV/c where both baryons and mesons have sufficient statistics.



(a) Ratio of Λ to K^0_S at multiplicities 80-100%.



(b) Ratio of Λ to K^0_S at multiplicities 0-20%.

Figure 19: Ratio of Λ to K^0_S at multiplicities 80-100% (left) and 0-20% (right).

The Λ/K^0_S ratio exhibits clear trends across multiplicity and transverse momentum. In the low-multiplicity events (80-100%), Junctions and Monash points show substantial scatter between points across the p_T range. Both models cluster around 0.65-0.75, and show a general rise-decline pattern. However, at $p_T > 6$ GeV/c both models show an upward tail approaching unity. This tail structure is puzzling and warrants further investigation, potentially informing our understanding of soft hadronisation in the low-multiplicity regime. A possible explanation is that high-p_T production originates from isolated hard scattering jets creating locally dense regions and enhanced baryons, but requires further investigation to clarify. In contrast, the high-multiplicity region shows no comparable tail. In the high-multiplicity panel the curve exhibits much smoother and more coherent behaviour, clustering much closer to unity (around 0.8-1.0) throughout the spectrum.

Most significantly, Junctions sits consistently above Monash ratios throughout the entire p_T range in the 0-20% spectrum, and the entire ratio curve shifts upwards. This indicates that junction mechanisms enhance Λ production relative to K^0_S in high-activity collisions. This enhancement is most dramatic during the intermediate p_T phase, at approximately $p_T \approx 2.5$ GeV/c, where the Junctions ratio reaches 1.4 compared to Monash's 1. At $p_T > 7$ GeV/c, the enhancement diminishes significantly, with both models converging towards 0.8. This p_T dependence suggests that junction mechanisms enhance baryons most optimally in the intermediate-p_T regime, where hadronisation transitions from soft to hard QCD. The overall smoother appearance of the high-multiplicity data, despite lower event statistics, may reflect that complex events contain multiple hard scatterings producing more abundant high-p_T particles, yielding better statistical distributions when integrated across many events.

The ALICE data provides further context to interpret the behaviour of models relative to experimental data. In the low-multiplicity regime ALICE shows a rise-peak-decline pattern, with a maximum at 0.9. At low-multiplicity, both models under-predict relative to ALICE, however from intermediate- to high-p_T ALICE data points sit relatively between both Monash and Junctions predictions. Trends become much clearer in the high-multiplicity (0-20%) bin, where the ALICE curve rises from 0.4 to peak at 1.2 around $p_T \approx 2.5$ GeV/c, and steadily declines toward 0.6. The ALICE curve lies between Monash and Junctions predictions in this percentile, tracking remarkably well with both models. At low-p_T and its peak it tends close to Junctions, and in the high-p_T region it tends closer to Monash, stretching below both models.

To conclude the analysis of the Λ/K^0_S ratio, a Chi-squared (χ^2) goodness-of-fit test was performed, allowing for quantitative comparison between the models - Monash and Junctions - and ALICE. The χ^2 value is calculated as per the formula [32]:

$$\chi^2 = \sum_i \frac{(\mathrm{O}i - \mathrm{E}i)^2}{\sigma \mathrm{O}, i^2 + \sigma \mathrm{E}, i^2} \tag{9}$$

where O denotes the observed value (experimental) and E the expected value (theoretical).

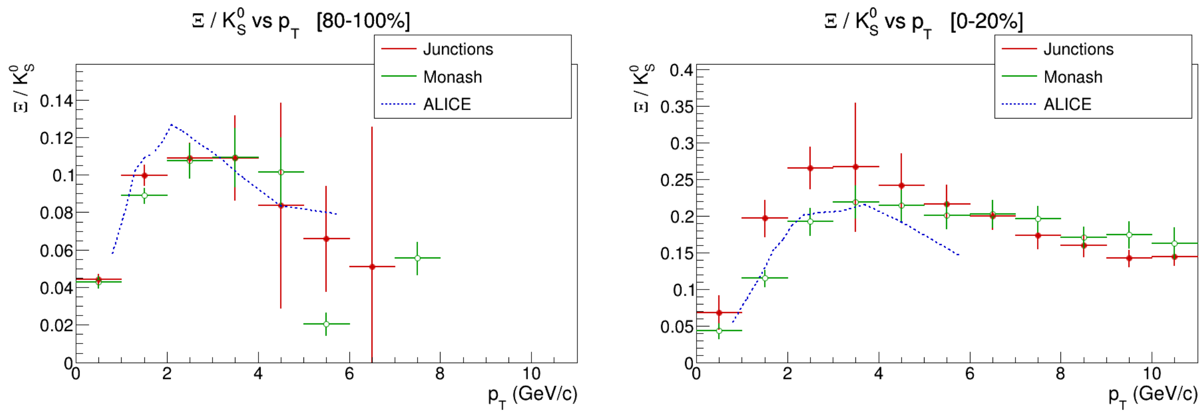
$$\chi^2 = \sum_i \frac{(\mathrm{ALICE}_i - \mathrm{PYTHIA}_i)^2}{\sigma^2_{\mathrm{ALICE},i} + \sigma^2_{\mathrm{PYTHIA},i}} \tag{10}$$

Lower values indicate better agreement between theory and experiment. NDF (number of degrees of freedom) is the number of p_T bins compared, and dividing χ^2 by NDF gives a normalised metric where values near unity indicate good agreement within uncertainties.

Table 5: Chi-squared goodness-of-fit for Λ/K^0_s ratio

Multiplicity Class	**Monash** χ^2/NDF	**Junctions** χ^2/NDF	**Better Agreement**	**Improvement Factor**	**Junctions Enhancement**
0-20%	6.51	2.74	Junctions	2.4×	+30.08%
20-40%	15.62	7.74	Junctions	2.0×	+8.55%
40-60%	15.59	7.96	Junctions	2.0×	−3.01%
60-80%	10.90	8.27	Junctions	1.3×	−9.91%
80-100%	6.77	2.99	Junctions	2.3×	+11.98%

The χ^2 analysis consistently demonstrates that Junctions outperforms Monash, across all multiplicity classes. The best agreement occurs in the outer most percentiles (80-100% and 0-20%). The *Junctions Enhancement* column quantifies the relative increase in baryon-to-meson production predicted in Junctions compared to Monash, averaged across all p_T bins. In the 0-20% bin Junctions predicts 30% more Λ/K^0_S than Monash, demonstrating strong junction effects at high-multiplicity. Neither model fully captures the strangeness dynamics in the intermediate percentiles. However, there are several limitations of the χ^2 analysis that must be considered. The combined error formula assumes PYTHIA and ALICE uncertainties are uncorrelated and Gaussian, which may not fully reflect the complex system. Secondly, the χ^2/NDF value exceeding unity may indicate not only model limitations but also underestimated uncertainties or unaccounted correlations in experimental data. Nevertheless, the consistent superiority of the Junctions tune proves that baryon junction formation plays a significant role in determining the Λ/K^0_S ratio.



(a) Ratio of Ξ to K^0_S at multiplicities 0-20%.

(b) Ratio of Ξ to K^0_S at multiplicities at 80-100%.

Figure 20: Ratio of Ξ to K^0_S at multiplicities at 80-100% (left) and 0-20% (right).

The Ξ/K^0_S ratio is significantly different from the Λ/K^0_S, due to the increased strangeness content (two strange quarks) of Ξ, and the rarity of the doubly-strange baryon. In the 80-100% plot, Junctions and Monash show substantial scatter, but the characteristic peak and decline pattern is still observed. The peak is located at $p_T \approx 3$ GeV/c with a ratio value of 0.13, and sits slightly above the predictions of both models. Both models show similar behaviour throughout this multiplicity bin, with no clear junction topology enhancement. Similarly to the previous plot, the 0-20% percentile exhibits a smoother, clearer curve. The entire curve is also similarly shifted up, with the ALICE peak shifted to a ratio value of 0.2. This confirms that high-multiplicity events produce more Ξ relative to K^0_S. In contrast to the previously discussed percentile, both Monash and Junctions exceed the ALICE peak, and in this intermediate-p_T region Junctions substantially enhances Ξ, but at high-p_T Monash surpasses Junctions.

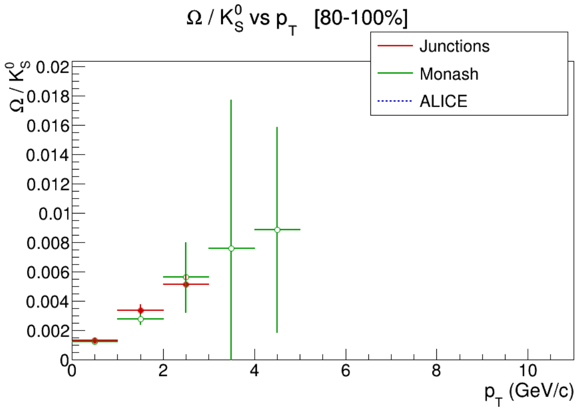
The ALICE data allows us to further interpret the performance of the models. In the 80-100% bin, and in the low-p_T region Junctions is slightly close to ALICE data points. After peaking, and steadily declining, moving from intermediate-p_T to high-p_T, ALICE sits between both models, reproduced slightly better than Junctions. However, it is difficult to conclude anything concrete from such minimal and sparse data points. In the 0-20% bin, ALICE data sits between both models in the initial p_T bins, with Monash outperforming Junctions. In the high-p_T tail region, where ALICE data is not available, Junctions converges toward the projected trend. However, limited ALICE data points in the high-p_T region restricts detailed analysis.

The same χ^2 test performed on the Λ/K^0_S ratios was repeated for the Ξ ratios. The analysis yielded the following results:

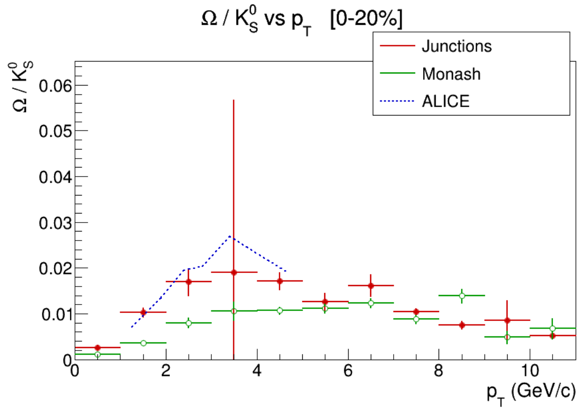
Table 6: Chi-squared goodness-of-fit for Ξ/K^0_s ratio

Multiplicity Class	**Monash** χ^2/NDF	**Junctions** χ^2/NDF	**Better Agreement**	**Improvement Factor**	**Junctions Enhancement**
0-20%	4.72	11.12	Monash	2.4×	+14.69%
20-40%	11.72	26.35	Monash	2.2×	+2.65%
40-60%	9.54	11.64	Monash	1.2×	−20.56%
60-80%	5.24	5.52	Monash	1.1×	+26.01%
80-100%	4.98	1.56	Junctions	3.2×	+36.44%

Notably, the χ^2 results of the Ξ/K^0_S ratio differ vastly from the Λ/K^0_S results. While the Λ/K^0_S ratio was reproduced well by Junctions across all multiplicity percentiles, the Ξ/K^0_S ratio is better reproduced by Monash with Junctions only outperforming Monash in the 80-100% range. At 0-20%, where junction enhancements should be most effective, Junctions overshoots ALICE data for most of the ratio curve. This suggests that Junctions potentially overproduces Ξ at higher multiplicities in proton-proton collisions, predicting more double-strange baryons than experimentally observed. Junctions parameters may require refinement for accurate multi-strange baryon production.



(a) Ratio of Ω to K_S^0 at multiplicities 80-100%.



(b) Ratio of Ω to K_S^0 at multiplicities 0-20%.

Figure 21: Ratio of Ω to K_S^0 at multiplicities at 80-100% (left) and 20-40% (right).

The Ω/K_S^0 ratio presents significant statistical challenges due to the extreme rarity of triple-strange baryons in proton-proton collisions. Ω particles, containing three strange quarks, are produced less frequently than both Λ and Ξ due to mass and strangeness content, resulting in sparse data with large uncertainties. The available measurements are illustrated in Figures 21a and 21b, but exhibit substantial scatter and lack clear trends. Despite statistical limitations the Ω ratio as per ALICE's experimental data, peaks at $p_T \approx 3.5$ GeV/c and a ratio value of 0.025. For the limited ALICE data points provided, Junctions best reproduces this line, with Monash positioned slightly below.
The Ω/K_S^0 χ^2 test must be interpreted with caution due to the aforementioned limitations of the dataset. ALICE data points are absent at multiplicity ranges 40-60% and 80-100%, nevertheless the table serves as supporting qualitative evidence.

Table 7: Chi-squared goodness-of-fit for Ω/K_s^0 ratio

Multiplicity Class	**Monash** χ^2/NDF	**Junctions** χ^2/NDF	**Better Agreement**	**Improvement Factor**	**Junctions Enhancement**
0-20%	28.66	3.86	Junctions	7.4×	+58.38%
20-40%	4.95	21.80	Monash	4.4×	+36.05%
40-60%	—	—	—	—	+1.85%
60-80%	73.51	65.39	Junctions	1.1×	−3.00%
80-100%	—	—	—	—	+6.43%

In the high-multiplicity regime, Junctions achieves remarkable agreement (χ^2/NDF = 3.86) with the ALICE data compared to Monash. This yields an improvement factor of 7.4, the largest observed across all three ratio analyses. This suggests Junctions preferentially produces even the rarest Ω baryons at high-multiplicity.

Detailed analysis of this triply-strange baryon would require much larger datasets for better interpretation.

3.4 Angular correlations

Angular correlations between Ξ baryons were analysed to investigate self-correlations of doubly-strange particles. In the previous section baryon-to-meson ratios, which quantify production, were considered. However, this section further explores the mechanics behind baryon-baryon interactions. Ξ-Ξ angular correlations are especially useful to probe the spatial and kinematic clustering of double-strange baryons, and whether Junctions produces Ξ pairs that cluster together more than Monash.

The two-dimensional correlation plots show the frequency of Ξ-pairs as a function of $\Delta\varphi$ and $\Delta\eta$.

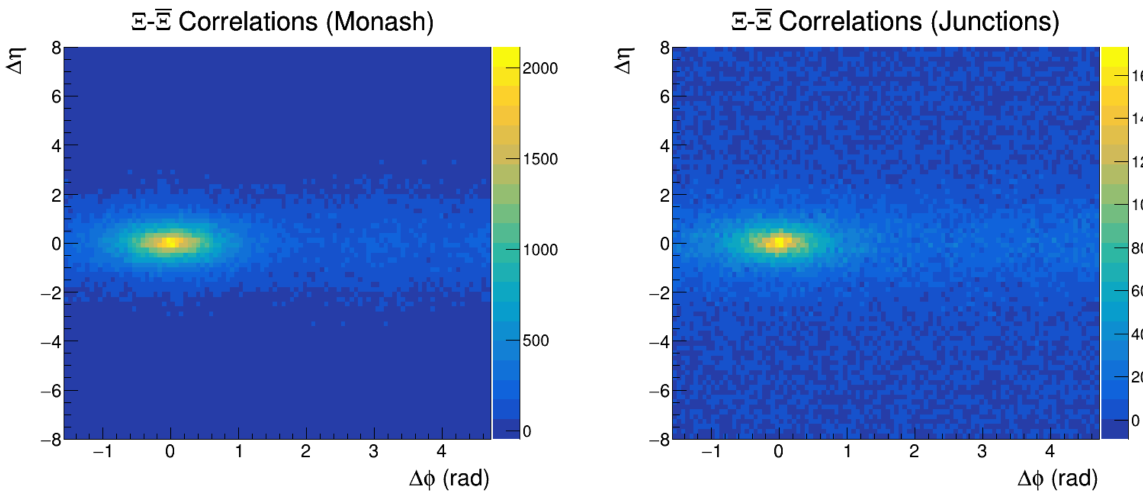


Figure 22: Ξ-Ξ 2D angular correlations using Monash (left) and Junctions (right).

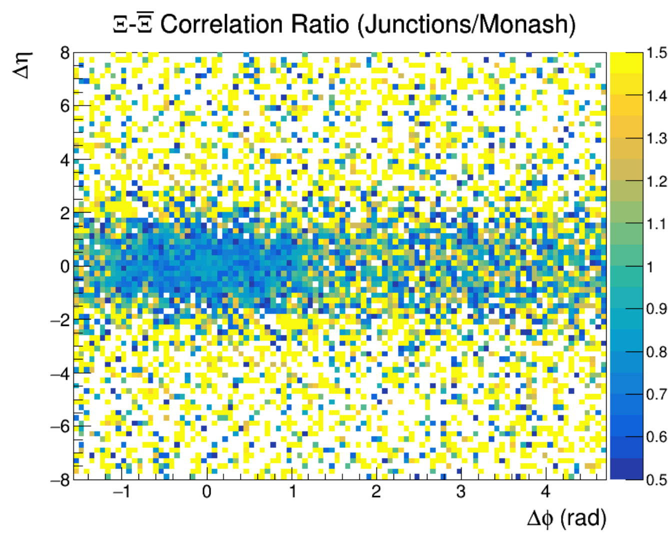
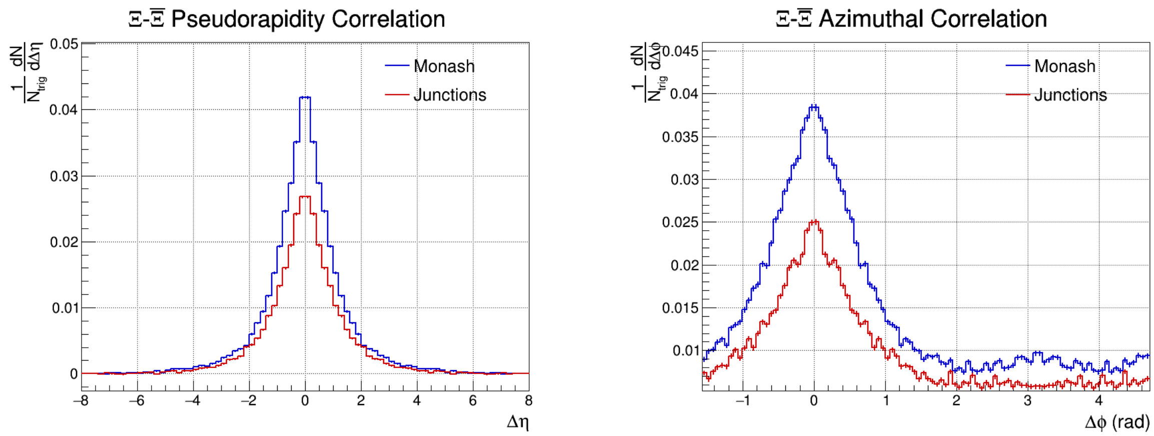


Figure 23: The ratio of Junctions/Monash of the 2D angular correlations plot.

Both Monash and Junctions graphs display similar peak structures: near-side peaks at $\Delta\varphi \approx 0$ and $\Delta\eta \approx 0$. This structure is consistent with Ξ pairs produced from same jet fragmentation. The Junctions/Monash ratio plot clearly shows blue at the near-side peak, indicating Junctions < Monash. Yellow, indicating enhancement and Junctions > Monash, dominates at larger angles. This indicates that Junctions produces lower correlation counts, which is surprising considering Junctions enhances overall Ξ yields in the baryon-to-meson ratio analysis. This paradox suggests that Junctions distribute these Ξ particles in a broader phase space. Note that the Junctions-Monash ratio displays significant scatter due to limited Ξ-Ξ pair statistics.

The Ξ-Ξ angular correlations projected into one-dimensional distributions in $\Delta\varphi$ and $\Delta\eta$ yield the following graphs.



(a) $\Delta\eta$ distribution (b) $\Delta\varphi$ distribution

Figure 24: 1D angular correlation plots: $\Delta\eta$ (left) and $\Delta\varphi$ (right).

Both Monash and Junctions exhibit near-side peaks in the one-dimensional projections. This demonstrates that Ξ-Ξ pairs preferentially have similar rapidity and azimuthal angles. Whilst both models exhibit the same structure, Monash has a peak amplitude exceeding Junctions. This distribution confirms that Junctions reduces correlation counts, but preserves the spatial distribution pattern. This suggests that junction-enhanced mechanisms reduce pair production while preserving the same underlying mechanics.

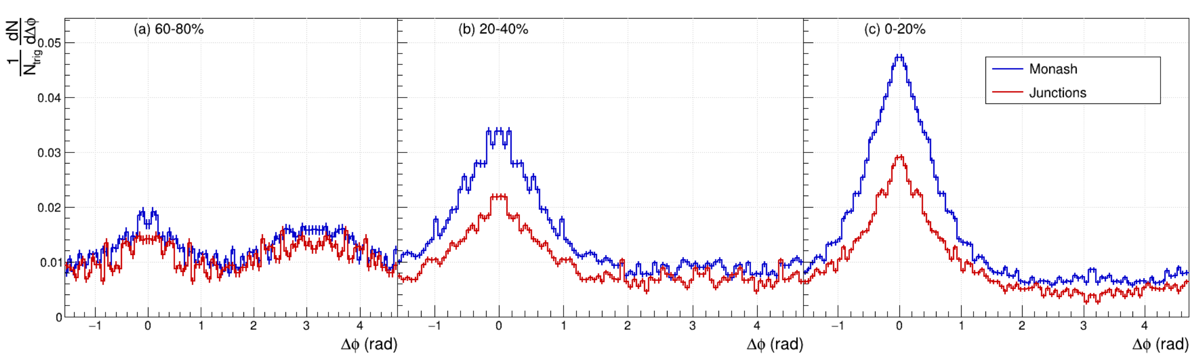


Figure 25: Ξ-Ξ azimuthal correlations as a function of event multiplicity: 60–80% (left), 20–40% (centre), 0–20% (right).

To conclude the analysis, Ξ-Ξ azimuthal correlations were examined as a function of event multiplicity. At low-multiplicity (60-80%), both models show flat distributions, with only slight peak-like features. At intermediate-multiplicity (20-40%), this peak becomes more pronounced, and at the highest multiplicity (0-20%) it is most prominent, and a clear near-side peak emerges. Across all multiplicity percentiles, Monash maintains a higher peak than Junctions. The progressive sharpening of the peak from low to high-multiplicity demonstrates that correlated Ξ pair production increases with event complexity.

4 Conclusion

The analysis begins with the characterisation of event multiplicity and particle yields of 100 million Monash generated events and 100 million Junctions generated events. The multiplicity distribution spans six orders of magnitude, and allows for percentile binning to be performed throughout the rest of the analysis. Particle yields scale strongly with multiplicity, with more exotic particles showing progressively larger enhancement factors. Junctions produces fewer mesons at fixed percentiles, consistent with CR mechanisms that preferentially create baryons. The enhancement factor of baryons exceeds unity for all species, confirming that junction topologies enhance baryon production. Critically, the enhancement factor scales with strangeness content — rare particles like Ω benefit most from Junctions mechanisms. Baryon-to-pion ratios exhibit clear upward trends with multiplicity, particularly pronounced for Junctions at high multiplicities. Both models significantly under-predict ALICE experimental yields, indicating fundamental limitations in current strange hadron reproduction.

Examination of the p_T-spectra across multiplicity classes reveals that both Junctions and Monash exhibit multiplicity-dependent spectral hardening — higher multiplicity events produce particles with higher p_T. This indicates a model-independent phenomenon. However, species' spectral shapes differ by strangeness content. For K_s^0 mesons, Junctions and Monash produce nearly identical spectra with only marginal differences at intermediate p_T. For Λ, Junctions produces softer p_T spectra with enhanced low-p_T production. This suggests junction topologies preferentially form single-strange baryons at lower p_T. For Ξ, Junctions shows a peak shift to higher p_T compared to Monash, indicating Junctions preferentially form double-strange baryons at higher p_T. For Ω, this trend continues even more dramatically, with a larger peak shift. This demonstrates a strangeness-dependent pattern: Junctions preferentially produce heavier, multi-strange baryons at higher p_T, while enhancing lighter baryons at lower p_T.

The baryon-to-meson ratios reveal the quantitative impact of junction enhancements. Λ/K_S^0 ratios demonstrate Junctions' effectiveness: Junctions outperforms Monash across all multiplicity percentiles and achieves superior agreement with ALICE throughout the spectrum (particularly at high multiplicities). The Ξ/K_S^0 ratios reveal limitations: Junctions over-predicts doubly-strange baryons relative to ALICE, and Monash achieves superior agreement. This suggests that Junctions requires refinement of the probability tuning for multi-strange particles. The Ω/K_S^0 ratio, despite severely limited statistics, shows Junctions' remarkable agreement with ALICE at the highest multiplicities. This indicates that Junctions reproduces even the rarest triple-strange baryon well and implements enhanced baryon production effectively.

The final section of the analysis, angular correlations of Ξ pairs, gives insight into the spatial distributions of the Ξ particle. Both Monash and Junctions exhibit near-side peaks at $\Delta\varphi \approx 0$ and $\Delta\eta \approx 0$, illustrating that Ξ pairs tend to have similar azimuthal and rapidity coordinates. The peak shapes are similar between models, however Monash shows higher peak amplitudes, indicating more correlated pairs, despite Junctions over-prediction of Ξ particles explored in the baryon-to-meson ratios. This resolution lies in the distribution of Ξ baryons: junction mechanisms produce Ξ particles that are not concentrated in the near-side correlation cone, but are instead distributed more broadly in phase space. Multiplicity dependence reveals that low-multiplicity collisions show featureless distributions, while structures emerge progressively and sharpen at high-multiplicity. This indicates denser events produce more correlated pairs.

Collectively, these findings demonstrate that hadronisation mechanisms in high-multiplicity proton-proton collisions exhibit strangeness dependent behaviour. The ALICE collaboration reports observing that this phenomena is not exclusive to heavy-ion collisions, but emerges in any system with sufficient particle-density [29]. The progressive enhancement of baryon production with strangeness content, combined with multiplicity-dependent correlation strengthening, additionally shows that recombination mechanisms traditionally associated with heavy-ion collisions operate in proton-proton collisions. Junctions implements junction topologies that capture important aspects of physics: superior performance on single-strange baryons, effective production of ultra-rare species, and systematic redistribution of baryons across phase space. However, significant limitations remain. Both models under-predict ALICE measurements across observables, indicating fundamental deficiencies in current strangeness modelling. Junctions' over-prediction of Ξ hints at mistuned probabilities for doubly-strange particles. Neither standard string fragmentation (Monash) nor current junction-enhanced approaches (Junctions) fully capture the complex strangeness dynamics observed experimentally. In summary, Junctions reproduces strange baryon production more effectively than Monash (especially for single-strange and triple-strange baryons at high multiplicities), yet neither model successfully describes ALICE data across all observables.

Continued improvement of Monte-Carlo generators such as PYTHIA directly benefits domains even beyond particle physics: they are used in medical-physics domains and radiation modelling for space-missions. By contributing to the understanding of particle production, this project participates in the log-term refinement of tools supporting technological innovation and scientific literacy across disciplines. Future refinements of Monte Carlo generators should address several shortcomings of the current tunes. Firstly, refinement of junction probability tuning for multi-strange particles. Secondly, the under-prediction of both models suggests fragmentation function parameters or additional hadronisation parameters are required. Future research should focus on parameter optimisation to match ALICE's experimental yields across all multiplicities. Alternative generators could also be explored to determine if limitations are PYTHIA-specific. Momentum balance studies could also be further explored. Eventually, it will also be possible to leverage the ALICE3 upgrade, expected to provide improved statistics for rare particles, such as Ω. Many limitations were encountered during this research, and should be considered in future works. The analysis employed PYTHIA 8 simulations at $\sqrt{s} = 7$ TeV to match the published ALICE measurements, but limits extrapolation to higher LHC energies. Rare particle statistics and ALICE data gaps were other restricting factors encountered during this research.

5 Critical reflection

This four-month Bachelor Thesis Research (BTR) period has been challenging yet rewarding in ways I could not have anticipated. Over these three months, I have learned lessons about research practice, particle physics and personal resilience that will shape my professional development going forward.

A hard lesson was learnt early on: after generating 200 million events at 14 TeV, I realised that matching ALICE's 7 TeV measurements was essential to my comparison. This required regenerating the entire dataset. It illustrates an important principle: in data intensive research, the initial planning and pre-analysis plan is of critical importance. Managing a dataset as large as this one, with various variables is quite overwhelming. Early in the analysis, goal setting is of the utmost importance, and knowing before even starting how you want to approach the analysis. This is something I could improve upon in future works, optimising the available time with regards to the goals I have in mind.

Before this project I had not formally studied particle physics. This BTR served as an intensive introduction to the world of particle physics and large-scale data analysis. By Utilising ROOT/C++ for event analysis, using NIKHEF servers for event generation, and conducting statistical tests, I developed practical skills I will carry into future research. Beyond technical skills, exploring the world of hadronisation and QCD has been extremely interesting and has fostered an authentic interest in the world of particle physics.

This project coincided with personal circumstances due to which I returned to my home country, uncertain of whether to continue my research. The decision to persist, supported by understanding mentors, my supervisor and family, taught me that research is not merely an intellectual exercise but is built on community support and relationships. Completing this thesis has improved my technical abilities, but also awakened a genuine interest in particle physics and collaboration, and I am excited about where the road will lead to next.

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A Additional Figures

A.1 Model-to-ALICE ratio for particle yields

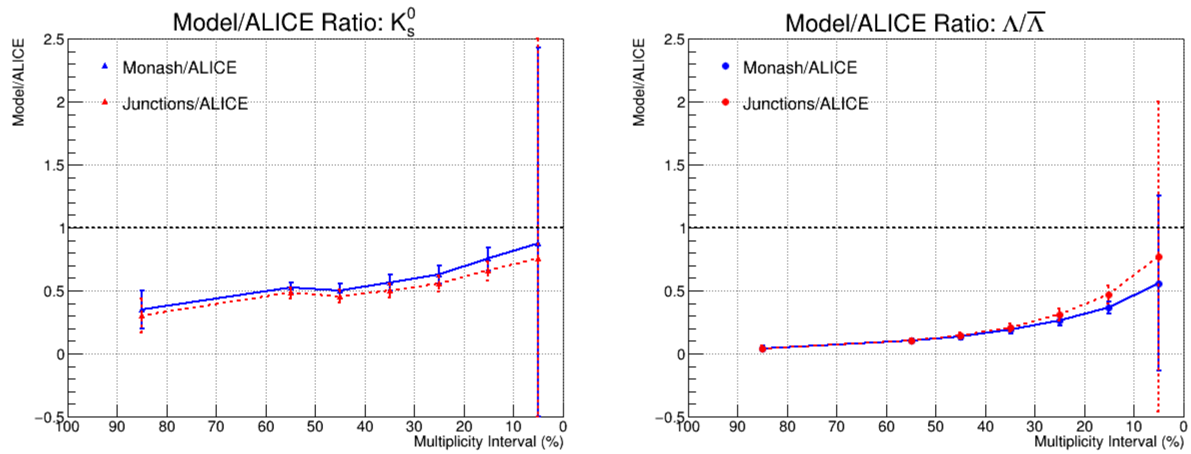


Figure 26: Ratio of simulation data from Monash and Junctions models to ALICE experimental data for K^0_S and Λ .

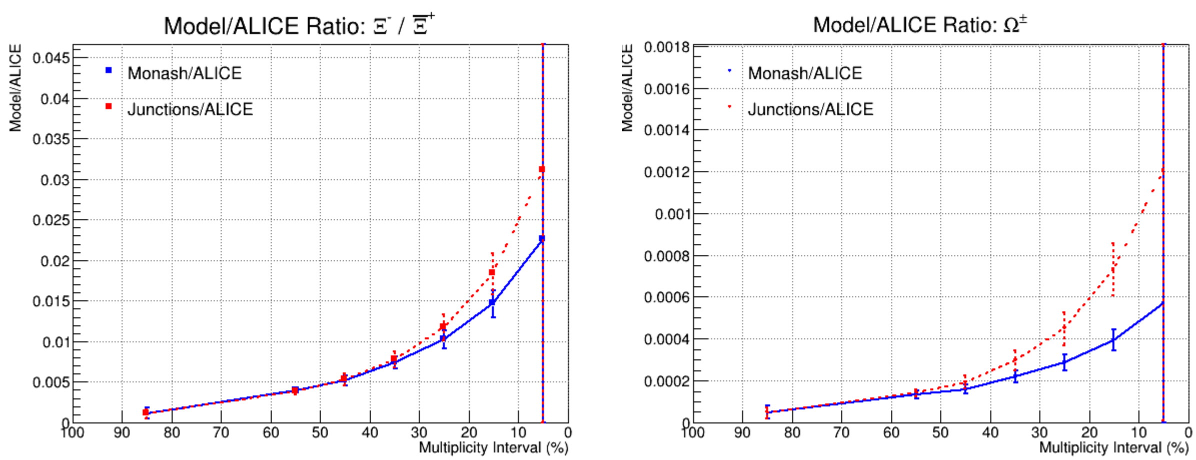


Figure 27: Ratio of simulation data from Monash and Junctions models to ALICE experimental data for Ξ and Ω.

A.2 Model-to-ALICE ratio for particle-to-pion yields

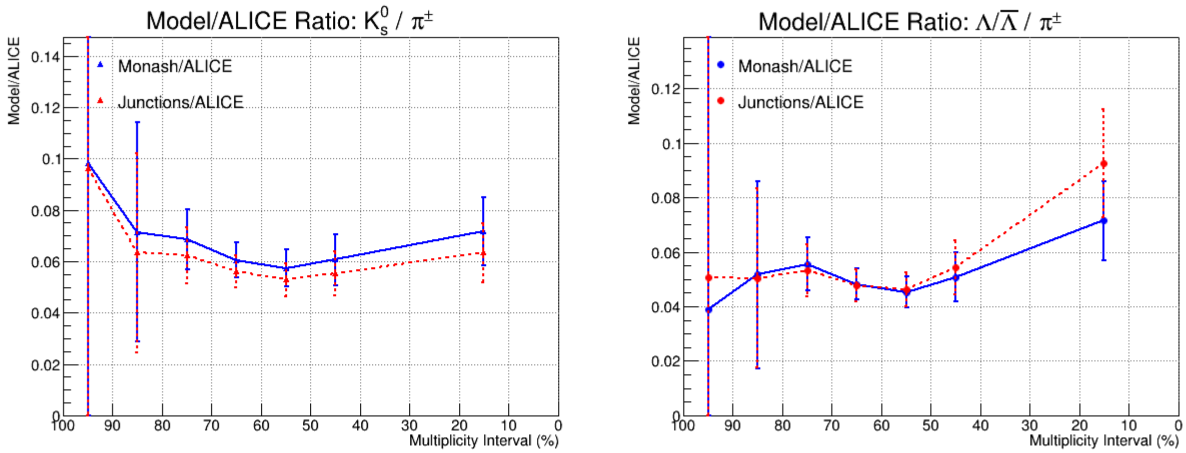


Figure 28: Ratio of simulation data from Monash and Junctions models to ALICE experimental data for K^0_S and Λ .

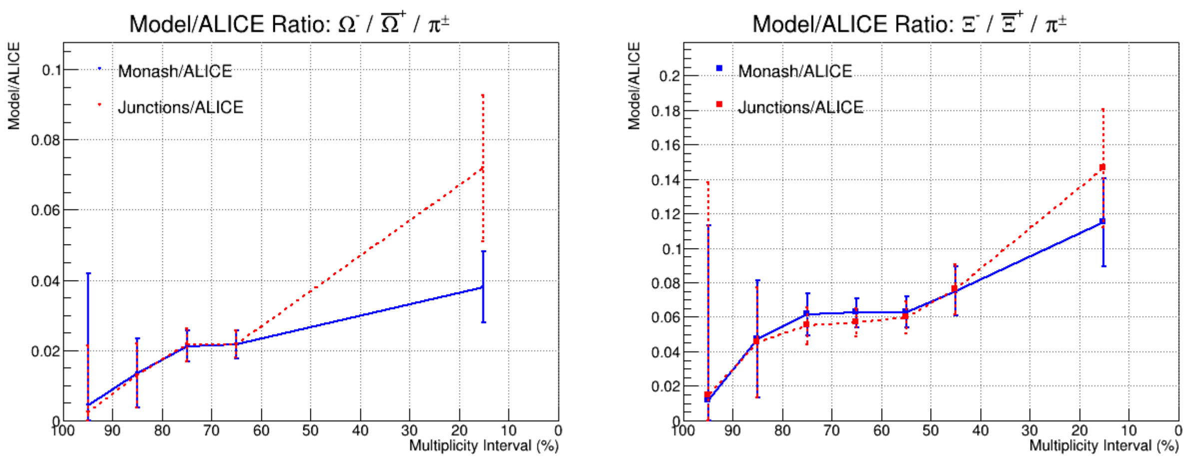
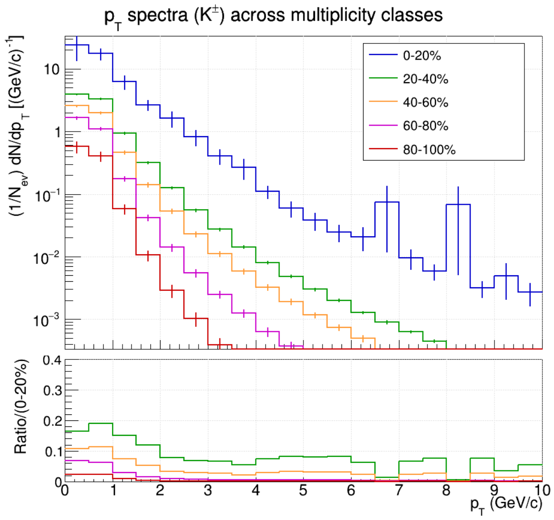
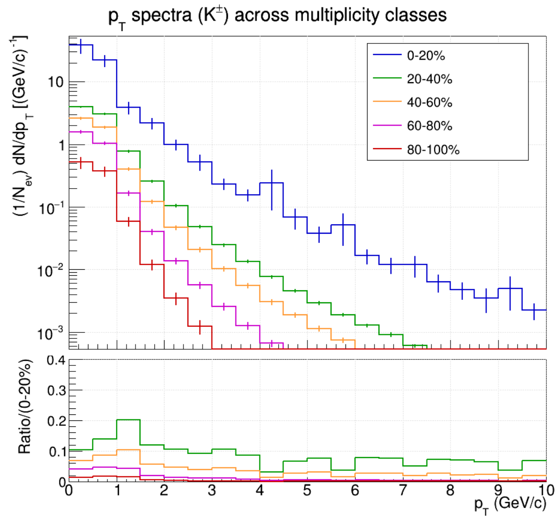
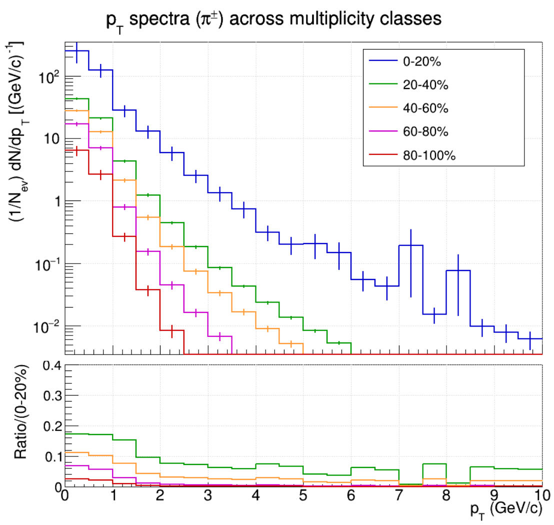


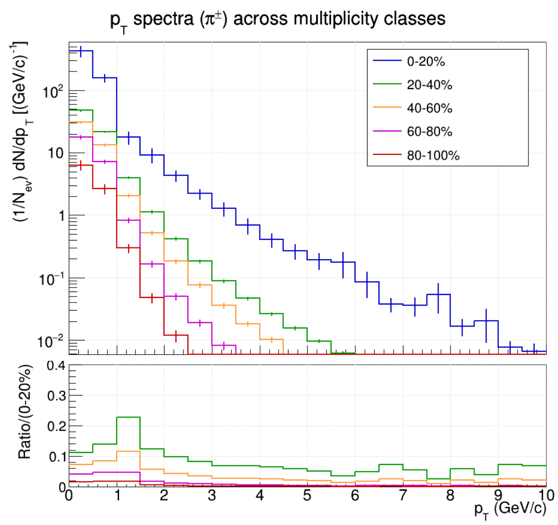
Figure 29: Ratio of simulation data from Monash and Junctions models to ALICE experimental data for Ω and Ξ.

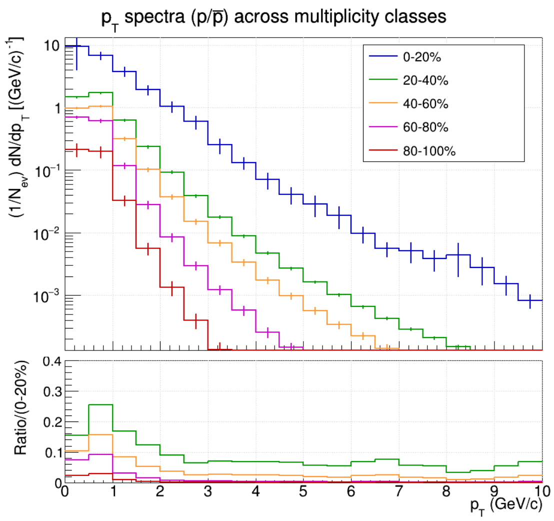
A.3 Pt Spectra: π, p, K



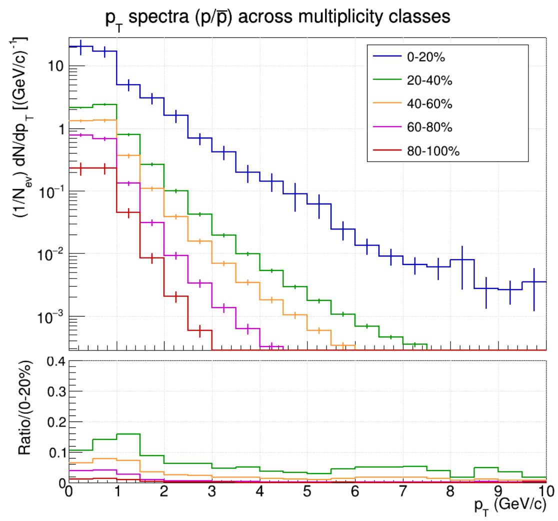








p_T spectra ($p/\bar{p}$) across multiplicity classes
$(1/N_{ev})\ dN/dp_T$ [(GeV/c)$^{-1}$]
0-20%
20-40%
40-60%
60-80%
80-100%
Ratio/(0-20%)
p_T (GeV/c)



p_T spectra ($p/\bar{p}$) across multiplicity classes
$(1/N_{ev})\ dN/dp_T$ [(GeV/c)$^{-1}$]
0-20%
20-40%
40-60%
60-80%
80-100%
Ratio/(0-20%)
p_T (GeV/c)